`

Question	517
Type	MSQ
Body	The input-output relationship between the system is $y(t) = x(\cos(3t))$. Then the system is
Option	Linear
Option	TIV
Option	Causal
Option	Dynamic
Correct	1,4
Explanation	$y(t) = x(\cos(3t))$ $y_1(t) = x_1[\cos(3t)]$ $y_2(t) = x_2[\cos(3t)]$ $y_3(t) = \alpha x_1(\cos(3t)) + \beta x_2(\cos(3t))$ $y_3(t) = \alpha y_1(t) + \beta y_2(t)$ So, linear system TIV/TV; $y(t) = x(\cos(3t))$ $y_1(t) = x[\cos(3t) - t_0]$ $y(t - t_0) = x[\cos(3(t - t_0))]$ $y_1(t) \neq y(t - t_0)$ TV $\Rightarrow y(t) = x[\cos(3t)]$ $y(0) = x(1)$ Present output depends on future input. so, non-causal and dynamic system.
Subject	Signals and Systems

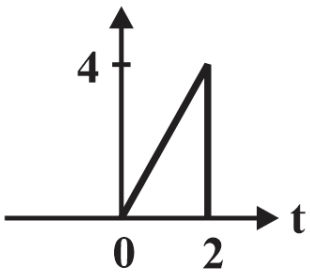
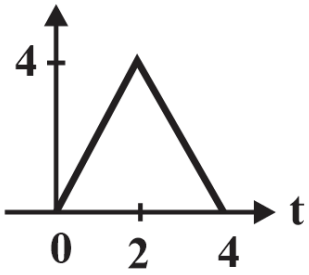
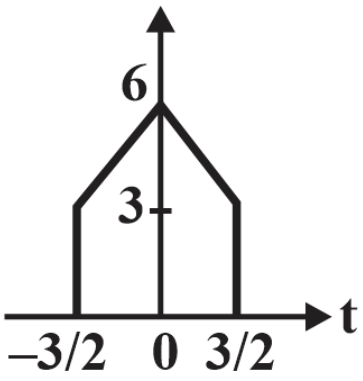
`

Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

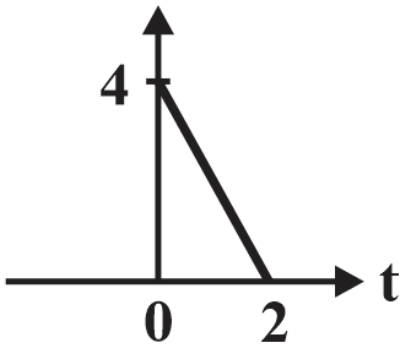
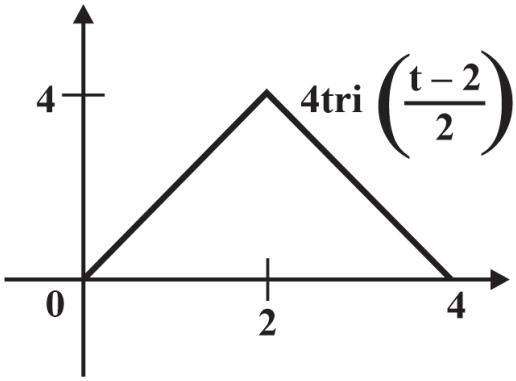
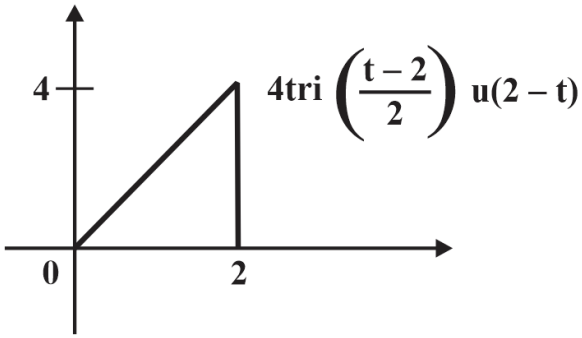
Question	518
Type	NAT
Body	The value of the summation $\sum_{n=0}^{\infty}(4)^n \delta(n+1)$ is _____
Correct	0
Explanation	From SIFTING property: $\sum_{n=N_1}^{N_2} x(n)\delta(n-n_0) = x(n_0)\ N_1 \le n \le N_2$ $= 0$ otherwise So, $\sum_{n=0}^{\infty}(4)^n \delta(n+1) = 0[n_0 = -1$, lies out side the limits$]$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

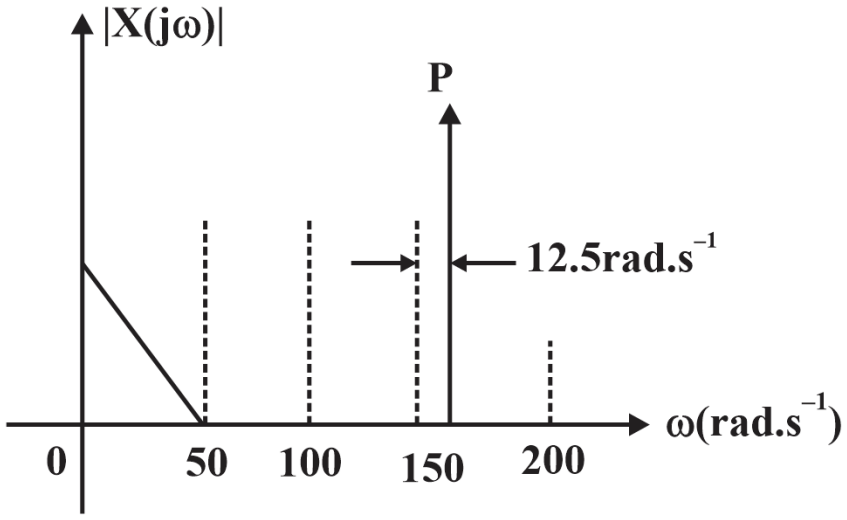
Question	519
Type	MCQ
Body	Which of the following graph corresponds to $x(t) = 4\text{tri}\left(\frac{t-2}{2}\right)u(2-t)$.
Option	
Option	
Option	

`

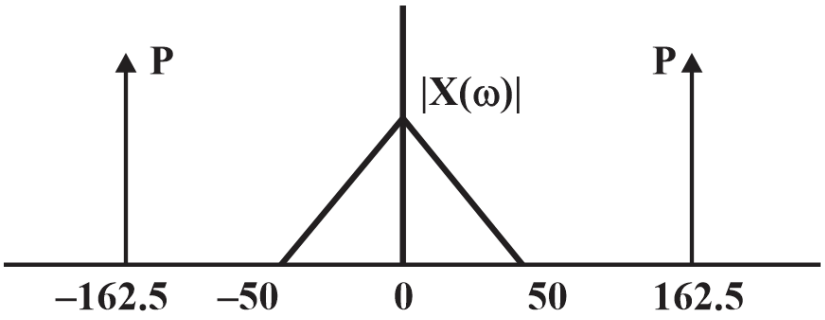
Option	
Correct	1
Explanation	$\mathrm{Tr}\left(\frac{t-\beta}{\alpha}\right)$ is a triangular pulse of width 2α centered at $t=\beta$  $u(2-t)=\begin{cases}1 & -\infty<t<2\\ 0 & \text{elsewhere}\end{cases}$ 
Subject	Signals and Systems
Topic	

`

Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	520
Type	MCQ
Body	$x(t)$ is a real continuous time signal whose magnitude frequency response $\left(\left\|X(j\omega)\right\|\right)$ is shown below. After sampling $x(t)$ at 100rad.s^{-1}, the spectral point P is down-converted to ____ rad.s^{-1} in the spectrum of the sampled signal. 
Option	12.5

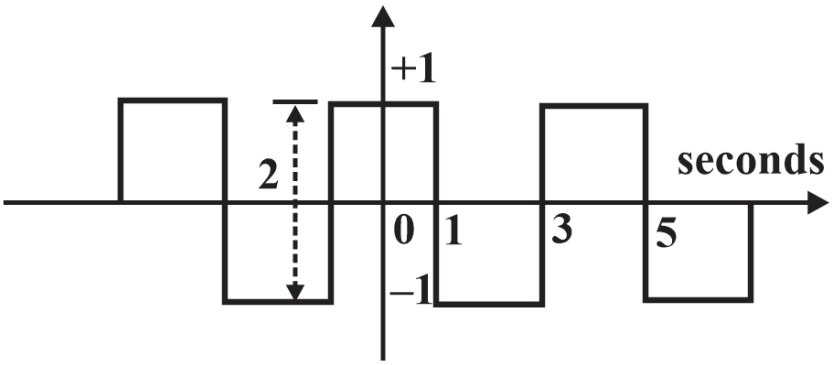
`

Option	25
Option	6.25
Option	37.5
Correct	4
Explanation	$\lvert X(\omega)\rvert$ is given for $\omega > 0$. We know $\lvert X(\omega)\rvert$ is even  $\omega_s = 100 \text{rad/sec}$ The F.T of sampled signal is $X_S(\omega) = \omega_s \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$ When the impulse at $\omega = -162.5$ is shifted right side by $2\omega_s = 200$, we can see an impulse at $200 - 162.5 = 37.5$. $\therefore$ The spectral point P is down converted to 37.5rad/sec.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	521
Type	MCQ
Body	Let $x(n)=\{16,8,4,2,1\}$. Then the signal $x(-n+4)$ is ____
Option	$\{0,1,2,4,8,16\}$
Option	$\{1,2,4,8,16,0\}$
Option	$\{1,2,0,4,8,16\}$
Option	$\{16,8,0,4,2,1\}$
Correct	1
Explanation	(1) $x(n) \rightarrow x(-n) \xrightarrow[\text{shiftby 4}]{\text{nigh}} x(4-n)x$ (2) $x(n) \xrightarrow[\text{by 4}]{\text{leftshift}} x[n+4] \xrightarrow{\text{Fold}} x[-n+4]$ $x(n+4)=\{16,8,4,2,1,\underset{\uparrow}{0}\}$ $x(-n+4)=\{\underset{\uparrow}{0},1,2,4,8,16\}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	522
Type	NAT
Body	The trigonometric Fourier series expansion of the periodic function in the figure has coefficients (a_n) and (b_n) for cosine and sine terms, respectively. The value of a_1 / a_3 is ____ . 
Correct	-3
Explanation	$a_n = \frac{2}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} x(t)\cos n\omega_0 t dt$ $x(t)$ is even $a_n = \frac{4}{T}\int_0^{\frac{T}{2}} x(t)\cos n\omega_0 t dt$ $T = 4$ $a_n = \frac{4}{4}\int_0^2 x(t)\cos n\omega_0 t dt$ $= \int_0^1 \cos n\omega_0 t dt + \int_1^2 -\cos n\omega_0 t dt$ $= \left(\frac{\sin n\omega_0 t}{n\omega_0}\right)_0^1 - \left(\frac{\sin n\omega_0 t}{n\omega_0}\right)_0^2$

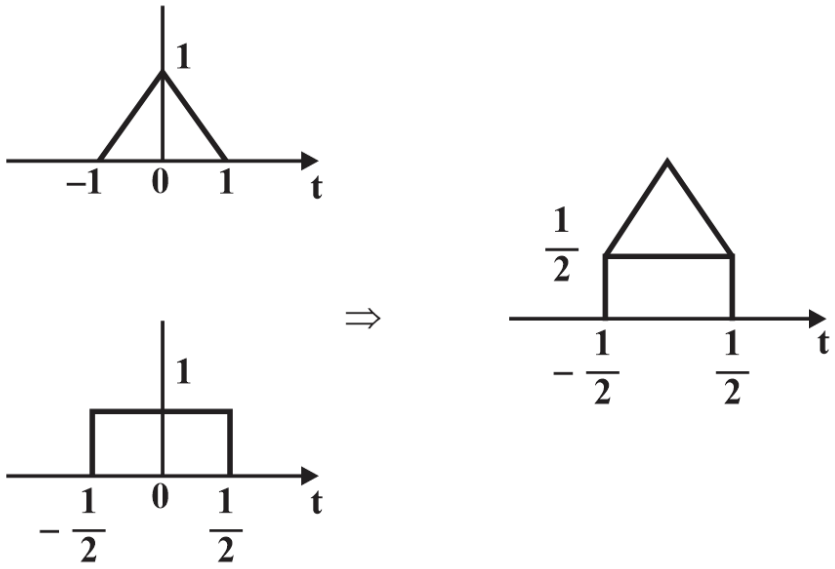
`

	$= \frac{1}{n\omega_0}\left[\sin n\omega_0 - \sin 0 - \sin 2n\omega_0 + \sin n\omega_0\right]$ $= \frac{1}{n\omega_0}\left[2\sin n\omega_0 - \sin 2n\omega_0\right]$ $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$ $a_1 = \frac{1}{(1)\frac{\pi}{2}}\left(2\sin\frac{\pi}{2} - \sin 2\left(\frac{\pi}{2}\right)\right)$ $a_1 = \frac{4}{\pi}$ $a_3 = \frac{1}{3\frac{\pi}{2}}\left(2\sin\frac{3\pi}{2} - \sin 3\pi\right)$ $= \frac{-4}{3\pi}$ $\frac{a_1}{a_3} = -3$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

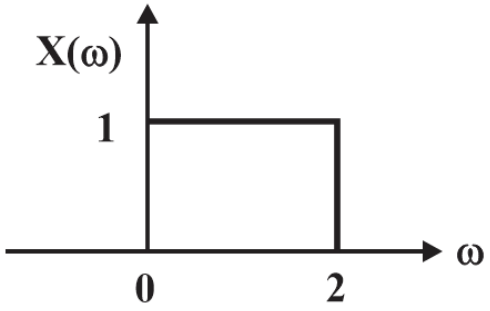
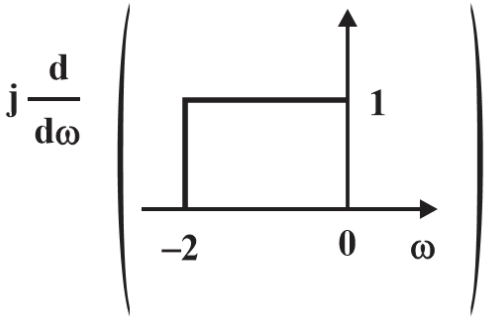
Question	523
Type	MCQ
Body	The value of the integral $\int_{-\infty}^{+\infty} \text{Sinc}^3(f)\,df$ is ____
Option	1
Option	3 / 4
Option	2 / 3
Option	1 / 2
Correct	2
Explanation	$\int_{-\infty}^{\infty} \text{Sin}\,c^2(f)\,\text{Sin}\,c(f)\,df = \int_{-\infty}^{\infty} \text{Tri}(t)\,\text{rect}(t)\,dt = \int_{-1/2}^{1/2} \text{Tri}(t)\,dt = 3/4 = 0.75$ OR Use Plancheral's theorem $\int_{-\infty}^{\infty} X(f)\,Y^*(f)\,df = \int_{-\infty}^{\infty} x(t)\,y(t)\,dt$ $= \int \text{Tr}(t)\,\text{rect}(t)\,dt$ $\text{Area} = \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}\right)(1)$ $= \frac{3}{4}$

`

	
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	524
Type	MCQ
Body	Let $X(\omega) = \text{rect}\left[\left(\frac{\omega - 1}{2}\right)\right]$ then Fourier transform of $tx(t)e^{-j2t}$ is ____

`

Option	$j\left[\delta(\omega+2)-\delta(\omega-2)\right]$
Option	$j\left[\delta(\omega+2)-\delta(\omega)\right]$
Option	$j\left[\delta(\omega)-\delta(\omega-2)\right]$
Option	$\text{rect}\left(\frac{\omega}{2}\right)\cdot e^{j2\omega}$
Correct	2
Explanation	 $tx(t)e^{-j2t}$ Apply F.T $[U\sin g - jtx(t) \leftrightarrow \frac{d}{d\omega}X(\omega)$ $\text{and}\, x(t)e^{j\omega_0 t} \leftrightarrow X(\omega-\omega_0)]$ $j\frac{d}{d\omega}\left[X(\omega+2)\right]$  $=j\left[\delta(\omega+2)-\delta(\omega)\right]$

`

Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	525
Type	MCQ
Body	Which of the following are odd functions? P: $\sin(t)$ $Q: \cos(t)$ R: $\sin(t)+\cos(t)$ S: $\sin(t)\cos(t)$
Option	Q and S
Option	P and Q
Option	P and S
Option	R and Q
Correct	3
Explanation	Let $f(t)=\text{sint}$ $f(-t)=\sin(-t)=-\text{sint}=-f(t)$

`

	$\Rightarrow$ sint is odd Let $g(t) = \cos t$ $g(-t) = \cos(-t) = \cos t = g(t)$ cost is even Let $h(t) = \sin t + \cos t$ $h(-t) = \sin(-t) + \cos(-t) = -\sin t + \cos t$ sint + cost is neither even nor odd Let $X(t) = \sin \cos t$ $x(-t) = \sin(-t)\cos(-t) = -\sin t \cos t = -x(t)$ $\Rightarrow$ sint cost is odd
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

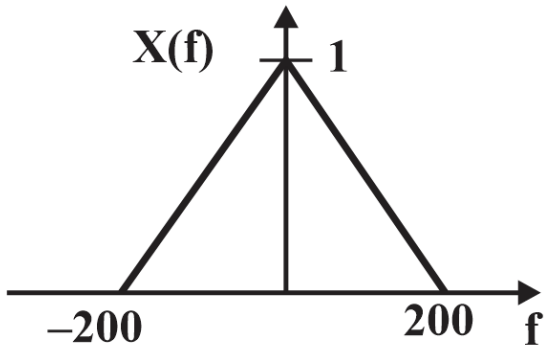
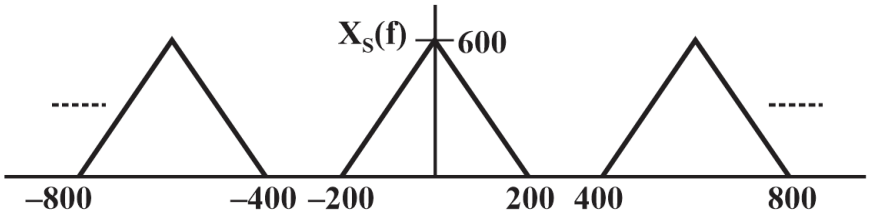
Question	526
Type	NAT
Body	If $x(n) = u(n) - u(n-5)$, and $h(n) = \delta(n) - \delta(n-1)$ and $y(n) = x(n) * h(n)$, then the value of $\sum_{n=-\infty}^{\infty} y(n)$ is _____ .

`

	(round off to the nearest integer)
Correct	0
Explanation	From the properties of convolution, we have $\sum_{n=-\infty}^{\infty} y(n) = \left(\sum_{n=-\infty}^{\infty} x(n) \right) \cdot \left(\sum_{n=-\infty}^{\infty} h(n) \right)$ $x(n) = \{1,1,1,1,1\}\, h(n) = \{1,-1\}$ $\sum_{n=-\infty}^{\infty} x(n) = 5, \sum_{n=-\infty}^{\infty} h(n) = 0$ $\therefore \sum_{n=-\infty}^{\infty} y(n) = 5 \times 0 = 0$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	527
Type	NAT
Body	A continuous time, band limited signal $x(t)$ has its Fourier transform described by

`

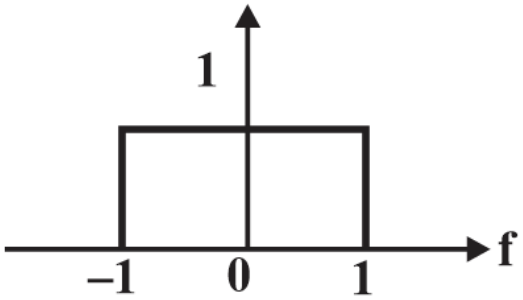
	$X(f)=\begin{cases}1-\dfrac{\lvert f\rvert}{200}, & \lvert f\rvert \le 200 \text{ Hz}\\ 0, & \lvert f\rvert > 200 \text{ Hz}\end{cases}$ The signal is uniformly sampled at a sampling rate of 600 Hz. The Fourier transform of the sampled signal is $X_S(f)$. What is the value of $\dfrac{X_s(600)}{X_s(500)}$?
Correct	2
Explanation	 The F.T of sampled signal is $X_S(f)=f_s\sum_{n=-\infty}^{\infty}X(f-nf_s)$  $X_S(600)=600$ $X_S(500)=X_S(-100)=600\left(1-\dfrac{\lvert-100\rvert}{200}\right)=300$ $\dfrac{X_S(600)}{X_S(500)}=\dfrac{600}{300}=2$
Subject	Signals and Systems

`

Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	528
Type	MCQ
Body	The Fourier transform of $u\left[1-\left\|\ t\right]\right.$ is ____
Option	$\mathrm{Sinc}\left(f\right)$
Option	$\frac{jd}{d\omega}\left[\frac{1}{j\omega}+\pi\delta\left(\omega\right)\right]$
Option	$2\mathrm{Sinc}\left(2f\right)$
Option	none of these
Correct	3
Explanation	$u\left[1-\left\|\ t\right]\right. = 1\ ;1-\left\|t\right\| > 0$ $1 > \left\|t\right\|$ $\Rightarrow \left\|t\right\| < 1$

`

	 $u\left[1-\left\|\ t\right]=\text{rect}\left(f\ /\ 2\right)\right.$ $\text{rect}\left(\frac{t}{2}\right)\rightarrow 2\,\text{Sin}\,c\left(2f\right)$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	529
Type	MCQ
Body	The time delay between the peaks of the voltage signals $v_1\left(t\right)=2\cos\left(6t+60^\circ\right)$ and $v_2\left(t\right)=3\sin\left(6t\right)$ is ____ s.
Option	$300\pi\ /\ 360$
Option	$100\pi\ /\ 360$

`

Option	$50\pi / 360$
Option	$200\pi / 360$
Correct	3
Explanation	$V_1(t) = 2\cos\left(6t + 60^\circ\right)$ $= 2\cos\left(6t + \frac{\pi}{3}\right)$ $= 2\sin\left(6t + \frac{\pi}{3} + \frac{\pi}{2}\right) \quad \because \cos\theta = \sin\left(\theta + \frac{\pi}{2}\right)$ $= 2\sin\left(6t + \frac{5\pi}{6}\right)$ $V_1(t) = 2\sin\left(6\left(t + \frac{5\pi}{36}\right)\right)$ _____(1) $V_2(t) = 3\sin(6t)$ _____(2) Comparing (1) and (2) $\text{Time delay} = \frac{5\pi}{36} = \frac{50\pi}{360}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	530
Type	MCQ
Body	Consider an LTI system with input and output related through the equation $y(t)=\int_{-\infty}^{t} e^{-(t-\tau)}x(\tau-4)d\tau$ Let the impulse response of above system is cascaded with another system having impulse response $\delta(t-4)$. The overall impulse response system is
Option	$h(t)=e^{-(t-4)}u(t-4)$
Option	$h(t)=e^{-(t-7)}u(t-4)$
Option	$h(t)=e^{-(t-8)}u(t-8)$
Option	$h(t)=-e^{-(t-3)}u(t-3)$
Correct	3
Explanation	$y(t)=\int_{-\infty}^{t} e^{-(t-\tau)}x(\tau-4)d\tau$ If $x(t)=\delta(t)$, then $y(t)=h(t)$ $h(t)=\int_{-\infty}^{t} e^{-(t-t)}\delta(\tau-4)d\tau$ $\therefore \quad h(t)=e^{-(t-4)}u(t-4)$ When this system is cascaded with another system $h_2(t)$  $h(t)=h_1(t)*h_2(t)$ $\therefore h(t)=e^{-(t-4)}u(t-4)*\delta(t-4)$ We know that $x(t+\alpha)^{*}\delta(t+\beta)$ $=x(t+\alpha+\beta)$

`

	$h(t)=e^{-(t-4-4)}u(t-4-4)$ $h(t)=e^{-(t-8)}u(t-8)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	531
Type	MCQ
Body	The number of samples required to yield enough information to describe the signal $x(t)=8+3\cos(8\pi t)+9\sin(4\pi t)$ is ____
Option	3
Option	2
Option	5
Option	1
Correct	3

`

Explanation	$\omega_0 = 4\pi \Rightarrow T = \frac{1}{2}\text{secs}$ At the Nyquist rate we would have 4 samples. We must have an integer number of samples in one period, sampled above the Nyquist rate. ∴ We need 5 samples.
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	532
Type	MCQ
Body	The following facts are given about the signal $x(t)$ with Laplace transform $X(s)$. 1. $x(t)$ is real and even 2. $X(s)$ has four poles and no zeros in the finite s-plane 3. $X(t)$ has a pole at $s = \frac{1}{2}e^{\frac{j\pi}{4}}$ 4. $\int_{-\infty}^{\infty} x(t)dt = 4$ Then $X(s)$ is

`

Option	$\frac{1}{\left(s^2-\frac{s}{\sqrt{2}}+\frac{1}{4}\right)\left(s^2+\frac{s}{\sqrt{2}}+\frac{1}{4}\right)}$
Option	$\frac{\frac{1}{4}}{\left(s^2+\frac{1}{4}\right)\left(s^2-\frac{1}{4}\right)}$
Option	$\frac{4}{\left(s^2-s+1\right)\left(s^2+s+1\right)}$
Option	$\frac{\frac{1}{4}}{\left(s^2-\frac{s}{\sqrt{2}}+\frac{1}{4}\right)\left(s^2+\frac{s}{\sqrt{2}}+\frac{1}{4}\right)}$
Correct	4
Explanation	Since $x(t)$ is real and even, the poles of $X(s)$ must occur in conjugate reciprocal pair, and symmetrical about jω axis. $X(s)=\frac{A}{\left(s-\frac{1}{2}e^{\frac{j\pi}{4}}\right)\left(s-\frac{1}{2}e^{-\frac{j\pi}{4}}\right)\left(s+\frac{1}{2}e^{-\frac{j\pi}{4}}\right)\left(s+\frac{1}{2}e^{\frac{j\pi}{4}}\right)}$ $X(s)=\frac{A}{\left(s^2-\frac{s}{\sqrt{2}}+\frac{1}{4}\right)\left(s^2+\frac{s}{\sqrt{2}}+\frac{1}{4}\right)}$ Also given $\int_{-\infty}^{\infty}x(t)dt=4$ So, $X(0)=4$ From equation (1)

`

	$X(0)=\frac{A}{\frac{1}{4}\times\frac{1}{4}}=4$ $A=4\times\frac{1}{16}=\frac{1}{4}$ So, $X(s)=\frac{\frac{1}{4}}{\left(s^2-\frac{s}{\sqrt{2}}+\frac{1}{4}\right)\left(s^2+\frac{s}{\sqrt{2}}+\frac{1}{4}\right)}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	533
Type	MCQ
Body	Find the ROC of $x(t)=e^{t}\sin(2t) \quad t<0$ $=0 \qquad t>0$
Option	$\sigma>-1$
Option	$\sigma<-1$

`

Option	$\sigma < 1$
Option	$\sigma > 1$
Correct	3
Explanation	$x(t) = e^{t} \sin(2t) u(-t)$ $e^{-t} \sin(2t) u(t) \leftrightarrow \frac{2}{(s+1)^2 + 4}, \sigma > -1$ $e^{t} \sin(2t) u(-t) \leftrightarrow \frac{-2}{(-s+1)^2 + 4}, \sigma < 1$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	534
Type	MCQ
Body	The unilateral Laplace transform of $x(t) = 2\delta(t+5) + 5\delta(t) + 3e^{-2(t+5)} u(t+4)$ is
Option	$2e^{5s} + 5 + \frac{3e^{-5(s+2)}}{s+2}$

`

Option	$2e^{5s}+5+\frac{3e^{-10}}{s+2}$
Option	$5+\frac{3e^{-10}}{s+2}$
Option	$5+\frac{3e^{-5(u+2)}}{s+2}$
Correct	3
Explanation	$x(t)=2\delta(t+5)+5\delta(t)+3e^{-2(t+5)}u(t+4)$ Unilateral L.T of $x(t)$ is $X(s)=\int_0^\infty x(t)e^{-st}dt=\int_0^\infty\left[5\delta(t)+3e^{-2(t+5)}\right]e^{-st}dt$ So, $X(s)=5+\frac{3}{s+2}e^{-10}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	535
Type	NAT

`

Body	The Inverse z-transform of $X(z)=\frac{1}{1024}\left[\frac{1024-z^{-10}}{1-\frac{1}{2}z^{-4}}\right],\lvert z\rvert>0$ is $x(n)$. The value of $x(3)$ is ____ (round off to three decimal places)
Correct	0.125
Explanation	The z-transform of $x(n)$ is $X(z)=\sum_{n=-\infty}^{\infty}x(n)z^{-n}$ If $x(n)=(\alpha)^n\ 0\le n\le N-1$ $=0$ otherwise Then $X(z)=\sum_{n=0}^{N-1}(\alpha)^n z^{-n}=\sum_{n=0}^{N-1}\left(\alpha z^{-1}\right)^n$ $X(z)=\frac{1-\left(\alpha z^{-1}\right)^N}{1-\alpha z^{-1}}$ $X(z)=\frac{1-\alpha^N z^{-N}}{1-\alpha z^{-1}}$ $X(z)=(\alpha)^N\left[\frac{\left(\frac{1}{\alpha}\right)^N-z^{-N}}{1-\alpha z^{-1}}\right]\lvert z\rvert>0$ _____(1) Given, $X(z)=\frac{1}{1024}\left[\frac{1024-z^{-10}}{1-\alpha z^{-1}}\right],\lvert z\rvert>0$ _____(2)

`

	Compare (1) and (2) So, $\alpha = \frac{1}{2},\ N = 10$ So, $x(n) = \left(\frac{1}{2}\right)^n \quad 0 \le n \le 9$ $= 0$ otherwise $x(3) = \frac{1}{8} = 0.125$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	536
Type	MCQ
Body	Let $x(n)$ be a signal whose z-transform is $x(z) = \frac{z}{z-2}, \lvert z \rvert > 2$ and given that $x_1(n) = \left(\frac{1}{4}\right)^n x(n)$. The ROC of $x_1(n)$ is
Option	$\lvert z \rvert > 2$

`

Option	$\left\|z\right\| > 8$
Option	$\left\|z\right\| > \frac{1}{2}$
Option	$\left\|z\right\| > 4$
Correct	3
Explanation	Given $X(z) = \frac{z}{z-2}, \left\|z\right\| > 2$ $x_1(n) = \left(\frac{1}{4}\right)^n x(n)$ Apply Z.T $X_1(z) = X\left(\frac{z}{\frac{1}{4}}\right) = X(4z)$ So, $X_1(z) = \frac{4z}{4z-2}, \left\|4z\right\| > 2$ $X_1(z) = \frac{z}{z-\frac{1}{2}}, \left\|z\right\| > \frac{1}{2}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

Question	537
Type	MCQ
Body	A feedback discrete system has a transfer function $H(z)=\frac{k}{1+k\left(\frac{z}{z-0.9}\right)}$. The range of "k" for system stability is
Option	$k<-0.1$
Option	$k<0.9$
Option	$k>-0.1 \& k<-1.9$
Option	For any positive value of k
Correct	3
Explanation	$H(z)=\frac{k(z-0.9)}{z-0.9+kz}=\frac{k(z-0.9)}{(1+k)z-0.9}$ $=\frac{k}{1+k}\left[\frac{z-0.9}{z-\frac{0.9}{1+k}}\right]$ For causality & stability $\left\|\frac{0.9}{1+k}\right\|<1$ (or) $0.9<\|1+k\|$ $k>-0.1 \& k<-1.9$
Subject	Signals and Systems

`

Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	538
Type	MCQ
Body	A finite impulse response (FIR) filter has only two non-zero samples in its impulse response $h(n)$, namely $h(0)=h(1)=1$. The Discrete Time Fourier Transform (DTFT) of $h(n)$ equals $H(e^{j\omega})$, as a function of the normalized angular frequency ω. For the range $\lvert\omega\rvert \le \pi, \lvert H(e^{j\omega})\rvert$ is equal to _____。
Option	$2\lvert\cos(\omega)\rvert$
Option	$2\lvert\sin(\omega)\rvert$
Option	$2\lvert\cos(\omega/2)\rvert$
Option	$2\lvert\sin(\omega/2)\rvert$
Correct	3

`

Explanation	$h(0)=1, h(1)=1$ $H(e^{j\omega})=\sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$ $=1\cdot e^{0}+1e^{-j\omega}$ $=1+e^{-j\omega}$ $=e^{-\frac{j\omega}{2}}\left(e^{\frac{j\omega}{2}}+e^{-\frac{j\omega}{2}}\right)$ $H(e^{j\omega})=2e^{-\frac{j\omega}{2}}\cos\frac{\omega}{2}$ $\left\|H(e^{j\omega})\right\|=\left\|2e^{-\frac{j\omega}{2}}\cos\frac{\omega}{2}\right\|=2\left\|\cos\frac{\omega}{2}\right\|$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	539
Type	MCQ

`

Body	Consider the 5 point averaging filter with impulse response $h(n)=\frac{1}{9}\{1,2,3,2,1\}$. The response of the filter to the input $(-1)^n$ is
Option	$(-1)^n$
Option	0
Option	$\frac{1}{9}(-1)^n$
Option	None
Correct	3
Explanation	$h(n)=\frac{1}{9}\{1,2,2,3,2,1\}$ $T.F.H(z)=\sum_{n=0}^{4}h(n)z^{-n}=\frac{1}{9}\left[1+2z^{-1}+3z^{-2}+2z^{-3}+z^{-4}\right]$ Input $x(n)=z^n \Rightarrow$ output $y(n)=z^nH(z)$ $=(-1)^n \Rightarrow y(n)=(-1)^nH(z)\|_{z=-1}$ $=\frac{1}{9}[1-2+3-2+1](-1)^n$ $=\frac{1}{9}(-1)^n$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

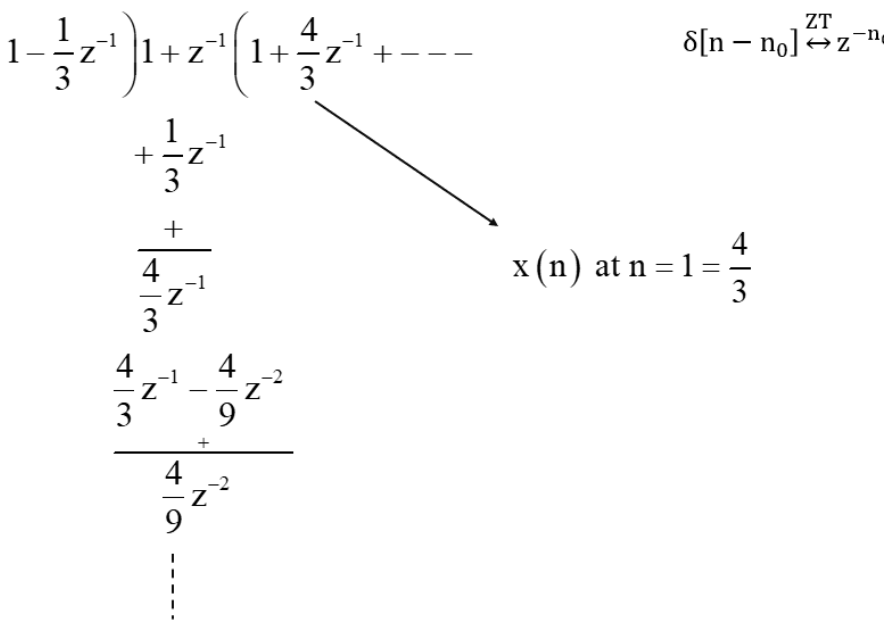
Question	540
Type	MCQ
Body	A linear system has Impulse response $h(t)=\left[\alpha+e^{-t}-e^{\beta t}\right]u(t)$, where α and β are real values. The system is stable if and only if
Option	$\alpha>0,\beta>0$
Option	$\alpha=0\,\&\,\beta<0$
Option	$\alpha=1/2,\beta>0$
Option	$\alpha=0\,\&\,\beta=2$
Correct	2
Explanation	$H(s)=\frac{\alpha}{s}+\frac{1}{s+1}-\frac{1}{s-\beta}$, if $\mathrm{Re}\{s\}>\mathrm{Re}\{\beta\}\,\beta\geq 0$ if $\mathrm{Re}\{s\}>0,\beta\leq 0$ For $\alpha=0$ and $\beta<0\Rightarrow j\omega$-axis be in the ROC of $H(s)$ for stability.
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-

`

Video Solution	-
PYQ	
OTS ID	

Question	541
Type	MCQ
Body	The Z-transform of a signal $x(n)$ is $X(z)=\frac{1+z^{-1}}{1-\frac{1}{3}z^{-1}};\lvert z\rvert>\frac{1}{3}$. Then the value of $x(n)$ at $n=1$ is ____
Option	$\frac{4}{3}$
Option	1
Option	$-\frac{4}{3}$
Option	$\frac{4}{9}$
Correct	1

`

Explanation	
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	542
Type	MCQ
Body	Consider the causal system $y(n)+y(n-1)+0.25y(n-2)=x(n)-2x(n-1).$

`

	This system is _____ .
Option	Minimum phase & stable
Option	Mixed phase & stable
Option	Minimum phase & unstable
Option	Mixed phase & unstable
Correct	2
Explanation	$H(z)=\frac{z^2-2z}{z^2+z+0.25}$ (z+0.5)² $\rightarrow$ Mixed phase (one zero at z = 2, z = 0) $\rightarrow$ Stable (2 poles at z=–0.5)
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	543
Type	NAT

`

Body	Let the DTFT of a signal $x(n)$ is $X(e^{j\omega})$ where $X(e^{j\omega}) = 5 + 6e^{-j2\omega}$. If another signal $y(n) = 5x(n+3)$, then the value of $Y(e^{j\omega})$ at $\omega = \pi$ is
Correct	–55
Explanation	$y(n) = 5x(n+3)$ $Y(e^{j\omega}) = 5e^{3j\omega}X(e^{j\omega})$ $Y(e^{j\omega}) = 5e^{j\omega}\left[5 + 6e^{-2j\omega}\right]$ $Y(e^{j\omega}) = 5\left[5e^{3j\omega} + 6e^{j\omega}\right]$ $Y(e^{j\pi}) = 5\left[5e^{3j\pi} + 6e^{j\pi}\right]$ $= 5[-5-6] = -55$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	544
Type	MCQ

`

Body	Consider $H(s)=\frac{3\,s^2+12\,s+48}{s^3+3\,s^2+4\,s+12};-3<\mathrm{Re}\{s\}<0.$ Then impulse response is _____
Option	$-3e^{-3t}u(t)+6\sin 2tu(-t)$
Option	$3e^{-3t}u(t)-6\sin 2tu(-t)$
Option	$6e^{-3t}u(t)-3\sin 2tu(-t)$
Option	$3e^{-3t}u(t)+6\sin 2tu(t)$
Correct	2
Explanation	$H(s)=\frac{3}{s+3}+\frac{12}{s^2+4};-3<\sigma<0$ $\downarrow_{\text{I.L.T}}$ $h(t)=3e^{-3t}u(t)-6\sin 2tu(-t)$ $\sin\omega_0 tu(t)\leftrightarrow\frac{\omega_0}{s^2+\omega_0^2};\sigma>0$ Applying L.T. $x(-t)\leftrightarrow x(-s)$ $\sin(-\omega_0 t)u(-t)\leftrightarrow\frac{\omega_0}{s^2+\omega_0^2};\sigma<0$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

Question	545
Type	MCQ
Body	A casual & stable LTI system has the property that input $x(n)=\left(\frac{2}{3}\right)^n u(n)$, output $y(n)=n\left(\frac{2}{3}\right)^n u(n)$. Determine the difference equation related to Input & Output.
Option	$y(n)-y(n-1)=x(n)$
Option	$y(n)-\frac{2}{3}y(n-1)=x(n)$
Option	$y(n)-\frac{2}{3}y(n-1)=\frac{2}{3}x(n-1)$
Option	$y(n)-\frac{2}{3}y(n-1)=\frac{2}{3}x(n)$
Correct	3
Explanation	$X(e^{j\omega})=\frac{1}{1-\frac{2}{3}e^{-j\omega}}, Y(e^{j\omega})=\frac{\frac{2}{3}e^{-j\omega}}{\left(1-\frac{2}{3}e^{-j\omega}\right)^2}$ $H(e^{j\omega})=\frac{Y(e^{j\omega})}{X(e^{j\omega})}=\frac{2/3e^{-j\omega}}{1-\frac{2}{3}e^{-j\omega}}$

`

	$Y\left(e^{j\omega}\right)-\frac{2}{3}e^{-j\omega}Y\left(e^{j\omega}\right)=\frac{2}{3}e^{-j\omega}X\left(e^{j\omega}\right)$ Apply IDTFT $y(n)-\frac{2}{3}y(n-1)=\frac{2}{3}x(n-1)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	546
Type	NAT
Body	The DFS coefficient of $x(n)=5+\sin\left(\frac{n\pi}{2}\right)+\cos\left(\frac{n\pi}{4}\right)$ is C_K. Then find the value of $\angle C_2+\angle C_1$ in radians
Correct	–1.57
Explanation	$\sin\left(\frac{n\pi}{2}\right)$ Period is $N_1=4$ $\cos\left(\frac{n\pi}{4}\right)$ period is $N_2=8$ The overall period is $N=8$

`

	$x(n)=5+\sin\left(\frac{n\pi}{2}\right)+\cos\left(\frac{n\pi}{4}\right)$ $x(n)=5+\frac{e^{\frac{jn\pi}{2}}-e^{-\frac{jn\pi}{2}}}{2j}+\frac{e^{\frac{jn\pi}{4}}+e^{-\frac{jn\pi}{4}}}{2}\quad -----(1)$ The Generalized expansion of $x(n)$ is $x(n)=\sum_{k=0}^{N-1}C_K e^{j\frac{2\pi}{N}nK}=\sum_{k=0}^{7}C_K e^{j\frac{2\pi}{8}nK}\quad -------(2)$ Compare (1) & (2) $C_0=5, C_2=\frac{1}{2j}, C_{-2}=\frac{-1}{2j}, C_1=\frac{1}{2}, C_{-1}=\frac{1}{2}$ $\angle C_2=\frac{-\pi}{2}, \angle C_1=0$ $\angle C_1+\angle C_2=\frac{-\pi}{2}=-1.57\,\text{radians}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	547
Type	MCQ

`

Body	The phase delay of the filter with I.R. $h(n)=\delta(n)-\delta(n-1)$ is
Option	$\frac{1}{2}$
Option	$-\frac{\pi}{2\omega}$
Option	$-\frac{\pi}{2\omega}+\frac{1}{2}$
Option	$-\frac{\pi}{2\omega}-\frac{1}{2}$
Correct	3
Explanation	$h(n)=\delta(n)-\delta(n-1)$ Applying F.T $H(e^{j\omega})=1-e^{-j\omega}=e^{-\frac{j\omega}{2}}\frac{\left[e^{\frac{j\omega}{2}}-e^{-\frac{j\omega}{2}}\right](2j)}{2j}$ $=2je^{-\frac{j\omega}{2}}\frac{\sin\omega}{2}$ Phase response $\theta(\omega)=-\frac{\omega}{2}+\frac{\pi}{2}$ $t_p(\omega)=-\frac{\theta(\omega)}{\omega}=\frac{1}{2}-\frac{\pi}{2\omega}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33

`

Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	548
Type	MCQ
Body	Let the input $x(n)=\{2,5,0,4\}$ produces the output as $y(n)=\{8,22,11,31,4,12\}$. Then Impulse Response is $h(n)$ is
Option	$\{1,3,4\}$
Option	$\{4,1,3\}$
Option	$\{1,3\}$
Option	$\{3,4,1\}$
Correct	2
Explanation	$x(n) \rightarrow m$ samples $= 4$ $h(n) \rightarrow N$ samples $y(n) \rightarrow m+N-1=6 \Rightarrow N=3$ $\quad 2a=8$ Assume $h(n)=\{a,b,c\}$ $\quad \Rightarrow a=4$ $5a+2b=22$ $x(n) \rightarrow 2 \quad 5 \quad 0 \quad 4$ $\quad 2b=2$ $h(n) \rightarrow a \quad b \quad c$ $\quad \Rightarrow b=1$ $2a \quad 5a \quad 0 \quad 4a$ $\quad 4c=12$

`

	$2b \quad 5b \quad 0 \quad 4b \qquad\qquad \Rightarrow c = 3$ $\qquad\qquad\quad 2c \quad 5c \quad 0 \quad 4c$ $y(n) \; 8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	549
Type	MCQ
Body	A system is described by Difference equation $y[n] = -0.5y(n-1) + x[n]$. Then, the output $y[n]$ value at $n = 0$ for the i/p $x[n] = \begin{cases} 1; & 0 \le n \le 2 \\ 0; & \text{elsewhere} \end{cases}$ is
Option	1
Option	0
Option	3
Option	0.5
Correct	1

`

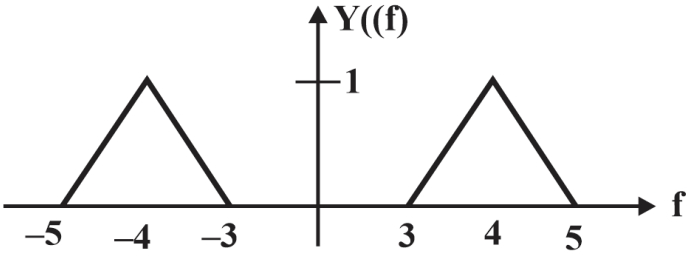
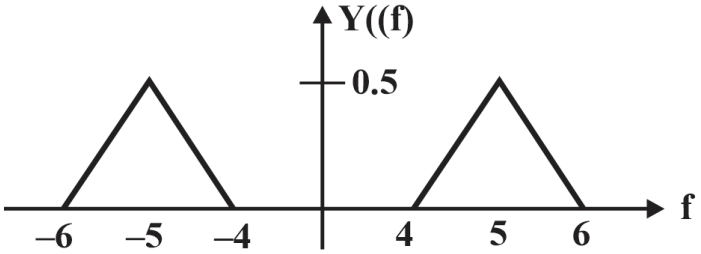
Explanation	$x[n]=\delta[n]+\delta[n-1]+\delta[n-2]$ $X(z)=1+z^{-1}+z^{-2}$ $Y(z)=X(z)H(z)$ $=\frac{1+z^{-1}+z^{-2}}{1+0.5z^{-1}}$ $=\frac{z^2+z+1}{z^2+0.5z}$ $y[n]=-0.5y[n-1]+x[n]$ By applying Z-transform $Y(z)=-0.5z^{-1}Y(z)+X(z)$ $H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1+0.5z^{-1}}$ $1+0.5z^{-1})1+z^{-1}+z^{-2}(1+0.5z^{-1}$ $\frac{1\pm 0.5z^{-1}}{0.5z^{-1}+z^{-2}}$ $\frac{0.5z^{-1}\pm 0.25z^{-2}}{}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

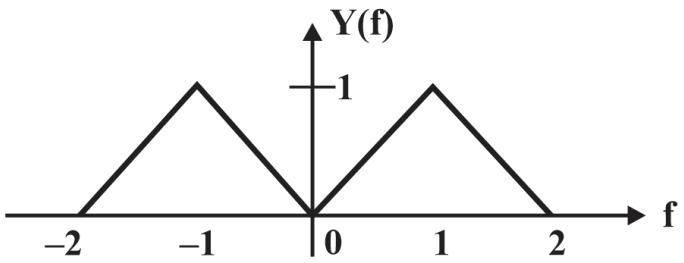
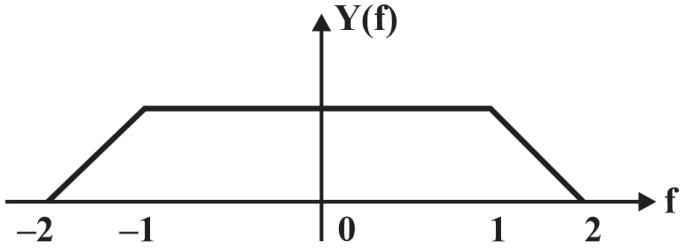
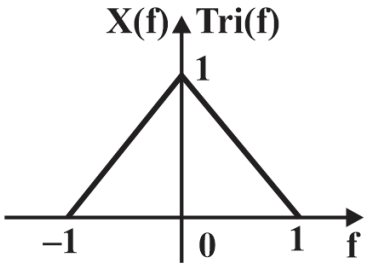
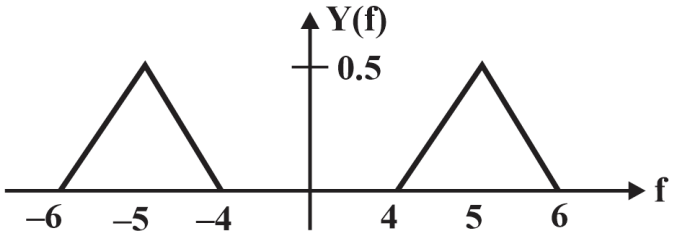
Question	550
Type	MCQ
Body	A system represented by difference equation $y[n]=0.25y[n-2]+x[n]$. Suppose the input is $x[n]=u[n]$, steady state output is _____
Option	$\frac{3}{4}$
Option	$\frac{4}{3}$
Option	0
Option	1
Correct	2
Explanation	$H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1-0.25z^{-2}}$ $x[n]=u[n]\rightarrow X(z)=\frac{1}{1-z^{-1}}$ $y(\infty)=Lt_{z\rightarrow 1}\left(1-z^{-1}\right)\underset{X(z)H(z)}{Y(z)}=\underset{z\rightarrow 1}{Lt}\frac{\left(1-z^{-1}\right)}{\left(1-z^{-1}\right)}\left[\frac{1}{1-0.25z^{-2}}\right]$ $=\frac{1}{1-0.25}=\frac{4}{3}$
Subject	Signals and Systems
Topic	
Correct Marks	2

`

Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	551
Type	MCQ
Body	The spectrum of modulated signal $y(t)=x(t)m(t)$ if $X(f)=Tr(f)$, where $Tr(f)=1;\|f\|\le 1$ and $m(t)=\cos(10\pi t)$ is
Option	
Option	

`

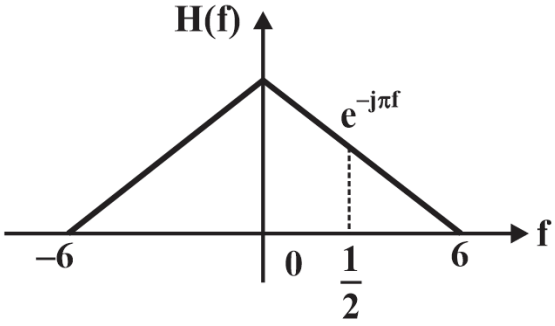
Option	
Option	
Correct	2
Explanation	$y(t)=x(t)\cos(10\pi t)=x(t)\left[\frac{e^{j10\pi t}+e^{-j10\pi t}}{2}\right]$ $x(t)e^{j2\pi f_c t}\leftrightarrow x(f-f_c)$  Apply Fourier transform $Y(f)=\frac{X(f-5)+X(f+5)}{2}$ 
Subject	Signals and Systems
Topic	

`

Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

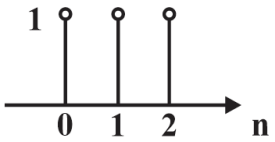
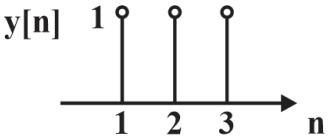
Question	552
Type	MCQ
Body	The frequency response of a filter is $H(f)=\text{Tri}\left(\frac{f}{6}\right)e^{-j\pi f}$. When we apply input as $\cos(\pi t)$, then output is
Option	$0.9167\cos(\pi t)$
Option	$0.9167\sin(\pi t)$
Option	$\cos(\pi t)$
Option	$-\cos(\pi t)$
Correct	2
Explanation	$H(f)=\text{Tri}\left(\frac{f}{6}\right)e^{-jxf}$

`

	 $x(t)=\cos(\pi t)$ $\omega_0=\pi \Rightarrow f_0=\frac{1}{2}\text{Hz}$ $\Rightarrow H(f_0)=\left(1-\frac{0.5}{6}\right)e^{-j\pi\left(\frac{1}{2}\right)}=0.9167\angle-\frac{\pi}{2}$ $y(t)=0.9167\cos\left(\pi t-\frac{\pi}{2}\right)=0.9167\sin(\pi t)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	553
Type	NAT

`

Body	The input - output of D.T L.T.I system are input $x[n]=u[n]-u[n\ -3]$, output $y[n]=u[n-1]-u[n-4]$. Then the Impulse response $h[n]$ at $n=1$ is ____ .
Correct	1
Explanation	$x[n]=u[n]-u[n-3]$ $y[n]=u[n-1]-u[n-4]$   $y[n]=x[n-1]$ $H(z)=z^{-1}$ By Inverse z-transform $h[n]=\delta[n-1]$ $\therefore h[n]\text{at}\,n=1\text{is}1$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	554
Type	NAT

`

Body	Given right sided signal $x(n)$ has Z.T $X(z)=\frac{1}{z^2-0.25}$. The value of $x(n)$ at $n=1$ is _____ .
Correct	0
Explanation	$X(z)=\frac{1}{z^2-0.25}=\frac{z^{-2}}{1-0.25z^{-2}}$ $1-0.25z^{-2})z^{-2}(z^{-2}+0.25z^{-4}+--\delta(n-n_0)\xleftrightarrow{Z.T}z^{-n_0}$ $\frac{z^{-2}-0.25z^{-4}}{+}\,0.25z^{-4}$ $0.25z^{-4}-0.0625z^{-6}$ Quotient doesn't have τ^{-1} term $\therefore x(n)$ at $n=1\Rightarrow 0$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	555
Type	MCQ

`

Body	Inverse Fourier transform of $Y(\omega)=\sin^2(3\omega)$ is ____
Option	$\frac{1}{4}\left[-\delta(t+6)+2\delta(t)-\delta(t-6)\right]$
Option	$\frac{1}{4}\left[\delta(t+6)+\delta(t)+\delta(t-6)\right]$
Option	$\delta(t)\sin 3t$
Option	$\frac{1}{4}\left[\delta(t+3)+\delta(t)-\delta(t-3)\right]$
Correct	1
Explanation	$Y(\omega)=\sin^2(3\omega)=\frac{1-\cos 6\omega}{2}=\frac{1}{2}-\frac{1}{2}\left[\frac{e^{j6\omega}+e^{-j6\omega}}{2}\right]$ $=\frac{1}{2}-\frac{1}{4}e^{j6\omega}-\frac{1}{4}e^{-j6\omega}$ $\boxed{\begin{array}{l}\delta(t)\xleftrightarrow{FT}1\\ \delta(t-t_0)\leftrightarrow e^{-j\omega t_0}\end{array}}$ Applying I.F.T $y(t)=\frac{1}{2}\delta(t)-\frac{1}{4}\delta(t+6)-\frac{1}{4}\delta(t-6)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	556
Type	MCQ
Body	A system S is described by the following differential equation $\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+x(t),\ x(0)=y(0)=0$ Where $x(t)$ and $y(t)$ are the input and output variables respectively. Let S_{INv} denotes inverse of S. If step signal is applied to the inverse system S_{INV} then the final value of the output is
Option	2
Option	4
Option	1
Option	0
Correct	1
Explanation	$\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+x(t)$ $SY(s)+2Y(s)=SX(s)+X(s)$ $\frac{Y(s)}{X(s)}=\frac{s+1}{s+2}=H(s)$ Transfer function of the inverse system S_{inv} is $H_{INV}(s)=\frac{1}{H(s)}=\frac{s+2}{s+1}$ $Y_{INV}=\frac{1}{s}H_{INV}(s)=\frac{1}{s}\left(\frac{s+2}{s+1}\right)$

`

	Final value $= \lim_{s\to 0} s\left(\frac{1}{s}\right)\left(\frac{s+2}{s+1}\right) = 2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	557
Type	MCQ
Body	The Laplace transform and ROC of the following signal $x(t) = e^{5t}u(-t+3)$ is
Option	$-\frac{e^{3(s+5)}}{s+5}, \sigma < -5$
Option	$-\frac{e^{-3(s-5)}}{s-5}, \sigma < 5$
Option	$\frac{e^{-3(s-5)}}{s-5}, \sigma < 5$
Option	$\frac{e^{3(s+5)}}{s+5}, \sigma < -5$

`

Correct	2
Explanation	$e^{-st}u(t) \xrightarrow{L.T} \frac{1}{s+5}, \sigma > -5$ $e^{-s(t+3)}u(t+3) \rightarrow \frac{e^{3s}}{s+5}, \sigma > -5$ $e^{-st}u(t+3) \rightarrow \frac{e^{3(s+5)}}{s+5}$ $e^{5t}u(-t+3) \rightarrow \frac{e^{3(-s+5)}}{-s+5} \sigma < 5$ $-\frac{e^{-3(s-s)}}{s-5}, \sigma < 5$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	558
Type	MCQ
Body	The value of $\int_{-5}^{5}\left(t^3+6t^2+2t+\cos\pi t\right)b''(t-2)dt$ is
Option	0

`

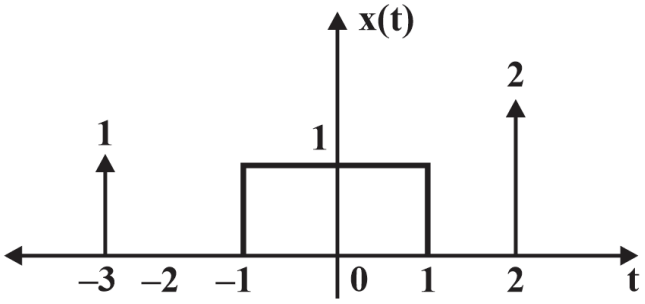
Option	$24-\pi^2$
Option	24
Option	$12-\pi^2$
Correct	2
Explanation	$\int_{-\infty}^{\infty} x(t)\delta''(t-t_0)dt=(-1)^2x''(t_0)$ Let $x(t)=t^3+6t^2+2t+\cos\pi t$ $x'(t)=3t^2+12t+2-\pi\sin\pi t$ $x''(t)=6t+12-\pi^2\cos\pi t$ $x''(2)=24-\pi^2$ $\int_{-5}^{5}(t^3+6t^2+2t+\cos\pi t)\delta''(t-2)dt=24-\pi^2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	559
Type	MCQ
Body	The Laplace transform of $x(t)=\int_0^t e^{-3\tau}\cos 2\tau\, d\tau$ is

`

Option	$\frac{s-3}{(s-3)^2+4}$
Option	$\frac{s-3}{s\left((s-3)^2+4\right)}$
Option	$\frac{s+3}{(s+3)^2+4}$
Option	$\frac{s+3}{s\left((s+3)^2+4\right)}$
Correct	4
Explanation	$\cos 2tu(t) \rightarrow \frac{s}{s^2+4}$ $e^{-3t}\cos 2tu(t) \rightarrow \frac{s+3}{(s+3)^2+4}$ $\int_0^t e^{-3\tau}\cos 2\tau d\tau \rightarrow \frac{s+3}{s\left((s+3)^2+4\right)}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	560
Type	NAT
Body	Consider the signal which is shown below.  The value of $\int_{-\infty}^{\infty} x(t)\,dt$ is _____ .
Correct	5
Explanation	The given signal consists of two impulses and a rectangular pulse $\int_{-\infty}^{\infty} x(t)\,dt = 1 + 2(1) + 2 = 5$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	561

`

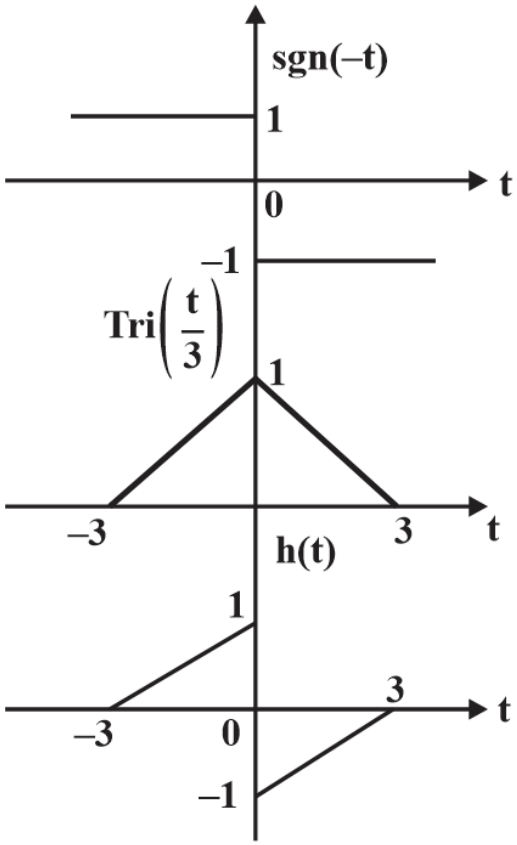
Type	MSQ
Body	Which of the following is/are even conjugate signals?
Option	e^{jt}
Option	je^{jt}
Option	$\frac{1}{a+jt}$
Option	te^{jt}
Correct	1,3
Explanation	Condition for even conjugate signal is $x^*(-t) = x(t)$ (a) $x(t) = e^{jt}, x(-t) = e^{-jt}, x^*(-t) = e^{jt}$ $\Rightarrow e^{jt}$ is even conjugate (b) $x(t) = je^{jt}, x(-t) = je^{-jt}, x^*(-t) = -je^{jt}$ $\Rightarrow je^{jt}$ is not even conjugate (c) $x(t) = \frac{1}{a+jt}, x(-t) = \frac{1}{a-jt} \cdot x^*(-t) = \frac{1}{a+jt}$ $\frac{1}{a+jt}$ is even conjugate (d) $x(t) = te^{jt}, x(-t) = -te^{-jt}, x^*(-t) = -te^{jt}$ te^{jt} is not even conjugate
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0

`

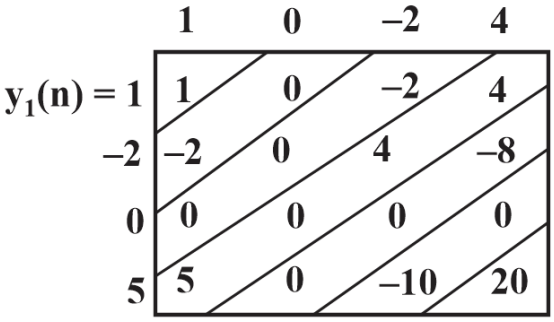
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	562
Type	MCQ
Body	The LTI system with impulse response $h(t)=\operatorname{sgn}(-t)\operatorname{Tr}\left(\frac{t}{3}\right)$ is
Option	causal and stable
Option	causal and unstable
Option	non causal and stable
Option	non causal and unstable
Correct	3

`

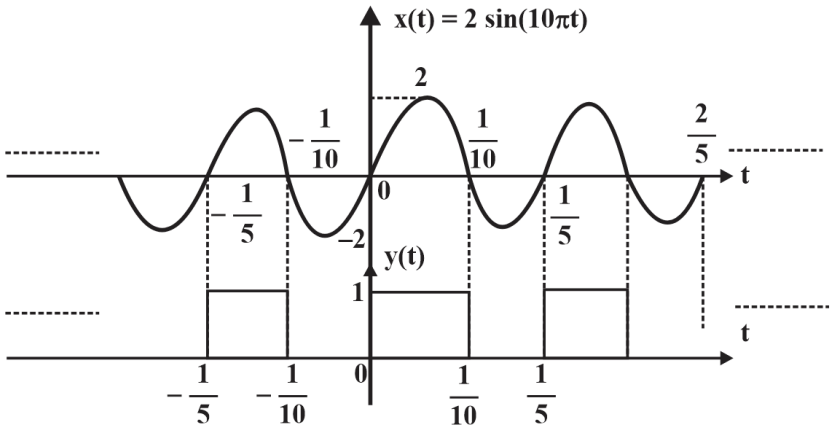
Explanation	 $h(t) \neq 0$ for $t < 0$, non-causal system. $\int_{-\infty}^{\infty} \lvert h(t) \rvert dt = \int_{-3}^{0} \lvert h(t) \rvert dt + \int_{0}^{3} \lvert h(t) \rvert dt < \infty$ So, stable system.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	563
Type	NAT
Body	Let $x(n)=\{1,-2,0,5\}$ and $h(n)=\{1,0,-2,4\}$ and $y(n)=x(n-1)*h(n+2)$. The value of $y(n)$ at $n=-3$ is ____ .
Correct	–2
Explanation	Assume $y_1(n)=x(n)*h(n)$  $y_1(n)=\{1,-2,-2,13,-8,-10,20\}$ Given $y(n)=x(n-1)*h(n+2)$ From shifting property of convolution $y(n)=y_1(n-1+2)=y_1(n+1)$ So, $y(n)=\{1,-2,-2,13,-8,-10,20\}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-

`

Video Solution	-
PYQ	
OTS ID	

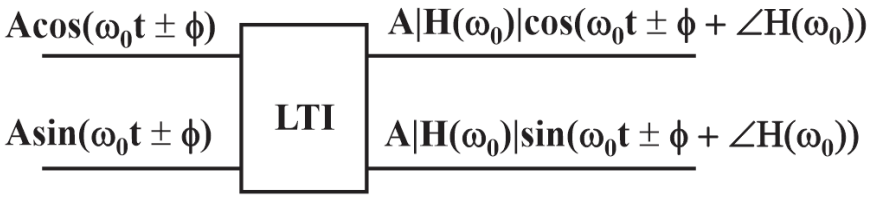
Question	564
Type	NAT
Body	Consider a system whose input and output related as $y(t) = u[x(t)]$. Where $x(t) = 2\sin(10\pi t)$. The power of the output signal is _____ .
Correct	0.5
Explanation	$u[x(t)] = 1 \quad x(t) > 0$ $= 0 \quad x(t) < 0$  Output is a periodic signal with $T_0 = \frac{1}{5}\text{sec}$ $p_y(t) = \frac{1}{T_0}\int_0^{T_0} \lvert y(t)\rvert^2 \, dt = \frac{1}{5}\int_0^{\frac{1}{5}} \lvert y(t)\rvert^2 \, dt$

`

	$p_y(t) = 5\int_0^{\frac{1}{10}} 1dt = \frac{1}{10} \times 5 = \frac{1}{2} = 0.5$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	565
Type	MSQ
Body	Consider the following input $x(t)$ and output $y(t)$ pairs for four different systems. Which of the following is/are LTI ?
Option	$x(t) = \cos(t), y(t) = \cos(t-1)$
Option	$x(t) = \sin(2t), y(t) = 5\sin\left(2t - \frac{\pi}{4}\right)$
Option	$x(t) = 2\cos(4t), y(t) = 4\sin\left(4t - \frac{\pi}{-}\right)$
Option	$x(t) = 5\sin(t), y(t) = 5\cos\left(2t - \frac{\pi}{3}\right)$

`

Correct	1,2,3
Explanation	If input to the LTI system is either cos(or) sine, then the output is either cos (or) sine with may be change in amplitude and phase, but the frequency is same 
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	566
Type	NAT
Body	The fundamental time period of $x(t)$ is 2sec. The fundamental time period of $y(t) = x(-2t+4)$ is _____ sec.
Correct	1
Explanation	$x(t) \rightarrow T_0 = 2\,\text{sec}$

`

	$x(t+4) \rightarrow T_0 = 2\,\text{sec}$ $x(2t+4) \rightarrow T_0 = 1\,\text{sec}$ $x(-2t+4) \rightarrow T_0 = 1\,\text{sec}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	567
Type	MCQ
Body	Consider the D.T signal $x(n) = \left(\frac{1}{4}\right)^n u(n)$. Define the signal $y(n) = x(-n)$. The value of $\sum_{n=-\infty}^{\infty} y(n)$ is
Option	$\frac{3}{4}$
Option	$\frac{4}{3}$
Option	$\frac{1}{3}$

`

Option	$\frac{2}{3}$
Correct	2
Explanation	$\left(\frac{1}{4}\right)^{n} u(n) \leftrightarrow \frac{1}{1-\frac{1}{4}e^{-j\omega}}$ $\left(\frac{1}{4}\right)^{-n} u(-n) \leftrightarrow \frac{1}{1-\frac{1}{4}e^{j\omega}}$ So, $y(n)=\left(\frac{1}{4}\right)^{-n} u(-n)$ $Y\left(e^{j\omega}\right)=\frac{1}{1-\frac{1}{4}e^{j\omega}}$ $Y\left(e^{j\omega}\right)=\sum_{n=-\infty}^{\infty} y(n)e^{-j\omega n}$ $Y\left(e^{j0}\right)=\sum_{n=-\infty}^{\infty} y(n)$ So, $\sum_{n=-\infty}^{\infty} y(n)=Y\left(e^{j0}\right)=\frac{1}{1-\frac{1}{4}}=\frac{4}{3}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

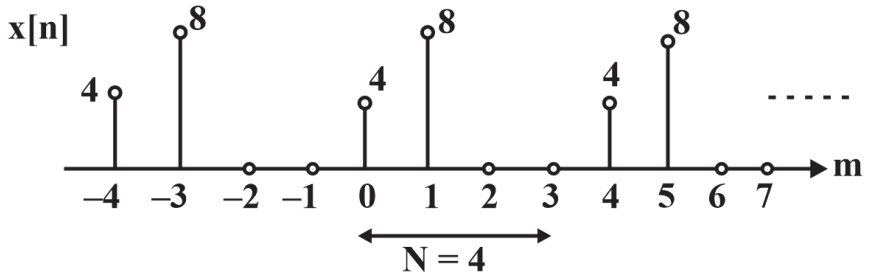
Question	568
Type	MCQ
Body	Consider two signals $x(n)=\left(\frac{1}{2}\right)^{n} u(-n+3)$ and $y(n)=\left(\frac{1}{2}\right)^{-n+2} u(n-1)$. The value of integral $\int_{0}^{2\pi} X\left(e^{j\omega}\right) Y\left(e^{-j\omega}\right) d\omega$ is
Option	$\frac{\pi}{2}$
Option	$\frac{\pi}{4}$
Option	$\frac{3\pi}{2}$
Option	$\frac{2\pi}{4}$
Correct	3
Explanation	From plan Cherel's relation $\sum_{n=-\infty}^{\infty} x(n) y^{*}(n)=\frac{1}{2\pi}\int_{0}^{2\pi} x\left(e^{j\omega}\right) Y^{*}\left(e^{j\omega}\right) d\omega$ Given $y(n)=\left(\frac{1}{3}\right)^{-n+2} u(n-1)$ is a real signal So, $Y^{*}\left(e^{j\omega}\right)=Y\left(e^{-j^{\omega}}\right)$

`

	So, $\frac{1}{2\pi}\int_0^{2\pi} X\left(e^{j\omega}\right) Y^*\left(e^{j\omega}\right) d\omega = \frac{1}{2\pi}\int_0^{2\pi} X\left(e^{j\omega}\right) Y\left(e^{-j\omega}\right) d\omega$ So, $\frac{1}{2\pi}\int_0^{2\pi} X\left(e^{j\omega}\right) Y\left(e^{-j\omega}\right) d\omega = \sum_{n=-\infty}^{\infty} x(n)\, y^*(n)$ So, $\frac{1}{2\pi}\int_0^{2\pi} X\left(e^{j\omega}\right) Y\left(e^{-j\omega}\right) d\omega = \sum_{n=-\infty}^{\infty} x(n)\, y^*(n)$ $\int_0^{2\pi} X\left(e^{j\omega}\right) Y\left(e^{-j\omega}\right) d\omega = 2\pi\left[\sum_{n=-\infty}^{\infty}\left(\frac{1}{2}\right)^n u(-n+3)\left(\frac{1}{2}\right)^{-n+2} u(n-1)\right]$ $= 2\pi\sum_{n=1}^{3}\left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^{-n+2} = 2\pi\sum_{n=1}^{3}\left(\frac{1}{2}\right)^2$ $= \frac{2\pi}{4}[3-1+1] = \frac{2\pi}{4}\times 3 = \frac{3\pi}{2}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

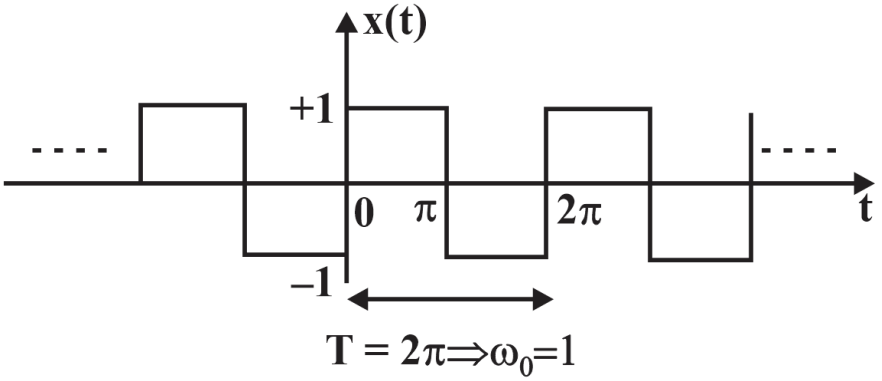
Question	569
Type	MCQ

`

Body	Consider the periodic signal $x[n]=\sum_{m=-\infty}^{+\infty}4\delta[n-4m]+8\delta[n-1-4m]$ having DFS coefficient C_k. Value of C_k is
Option	$1+2e^{-j\frac{k\pi}{2}}$
Option	$1-2e^{-j\frac{kz}{2}}$
Option	$1+e^{-jk\pi}$
Option	$1-2e^{-jkk}$
Correct	1
Explanation	$x[n]=\sum_{m=-\infty}^{+\infty}4\delta[n-4m]+8\delta[n-1-4m]$ $=\ldots\ldots+4\delta[n+4]+8\delta[n+3]+4\delta[n]+8\delta[n-1]+4\delta[n-4]+8\delta[n-5]+\ldots\ldots$  One period of $x[n]=\{4,8,0,0\}$ $C_k=\frac{1}{N}\sum_{n=0}^{N-1}x[n]e^{-j\frac{2\pi}{N}kn}=\frac{1}{4}\left[4+8e^{-j\frac{2\pi}{4}k}\right]=1+2e^{-j\frac{k\pi}{2}}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-

`

Video Solution	-
PYQ	
OTS ID	

Question	570
Type	MCQ
Body	A periodic signal $x(t)$ has fundamental frequency $\omega_0 = 1$ and one period of it is $x_1(t) = u(t) - 2u(t-\pi) + u(t-2\pi)$ with E.F.S coefficient C_n we can conclude that ____ .
Option	$C_0 \neq 0, C_n = 0$ for even " n "
Option	$C_0 = 0, C_n = 0$ for even " n "
Option	$C_0 \neq 0, C_n = 0$ for odd " n "
Option	$C_0 = 0, C_n = 0$ for odd " n "
Correct	2
Explanation	 Odd + H.W symmetry ⇒ C_n exists only for odd harmonics
Subject	Signals and Systems
Topic	

`

Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	571
Type	MCQ
Body	A real signal $x(t)$ whose EFS coefficient value at $n = 2$ is $\frac{1}{j\pi}$. Then the value of polar form of TFS coefficient d_n value at $n = 2$ is
Option	$\frac{2}{j\pi}$
Option	$\frac{2}{\pi}$
Option	$\frac{1}{\pi}$
Option	$\frac{1}{j\pi}$
Correct	1
Explanation	The relation between d_n and c_n is $d_n = 2c_n n \neq 0$

`

	$d_2 = 2c_2$ $d_2 = \frac{2}{j\pi}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	572
Type	MCQ
Body	DTFT of a real signal $x[n]$ is $X(e^{j\omega})$. DTFT of $e^{\pi n}x[n-1]$ is
Option	$e^{-j\omega}X(e^{j\omega})$
Option	$e^{-j(\omega-1)}X(e^{j(\omega-1)})$
Option	$-X[e^{j(\omega-z)}]e^{-j\omega}$
Option	$X(e^{j(\omega-\pi)})e^{j\omega}$

`

Correct	3
Explanation	$x[n-n_0] \xleftrightarrow{FT} e^{-j\omega n_0} X\left(e^{j\omega}\right)$ $x[n] e^{j\omega_c n} \leftrightarrow X\left[e^{j(\omega-\omega_c)}\right]$ $x[n-1] \xleftrightarrow{FT} e^{-j\omega} X\left(e^{j\omega}\right) \quad e^{j\pi} = -1$ $e^{j\pi n} x[n-1] \leftrightarrow e^{-j(\omega-\pi)} X\left[e^{j(\omega-\pi)}\right]$ $= -e^{-j\omega} X\left[e^{j(\omega-\pi)}\right]$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	

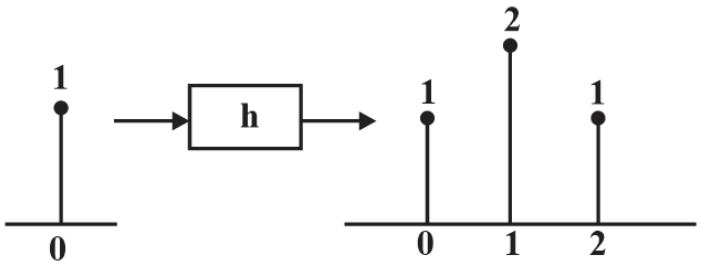
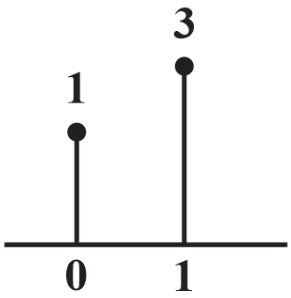
Question	573
Type	MCQ
Body	The Z.transform of nu(-n) is
Option	$\frac{z}{(z+1)^2}; \lvert z\rvert < 1$
Option	$\frac{-z}{(z-1)^2}; \lvert z\rvert < 1$

`

Option	$\frac{1}{(z-1)^2};\|z\|<1$
Option	$\frac{z}{(z+1)^2};\|z\|<1$
Correct	2
Explanation	$nu(n)\leftrightarrow\frac{z^{-1}}{\left(1-z^{-1}\right)^2};\|z\|>1$ apply T.R property $x(-n)\leftrightarrow X\left(z^{-1}\right),ROC=\frac{1}{R}$ Apply T.R property $-nu(-n)\leftrightarrow\frac{\left(z^{-1}\right)^{-1}}{(1-z)^2}$ $nu(-n)\leftrightarrow\frac{-z}{(1-z)^2}=-\frac{z}{(z-1)^2};\|z\|<1$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	574

`

Type	NAT
Body	The input-output relationship of an LTI system is given below.  For an input $x[n]$ shown below  The peak value of the output when $x[n]$ passes through h is ____ .
Correct	7
Explanation	$\delta(n) \rightarrow h(n) = \delta(n) + 2\delta(n-1) + \delta(n-2)$ $x(n) = \delta(n) + 3\delta(n-1) \rightarrow y(n) = h(n) + 3h(n-1)$ $y(n) = \delta(n) + 2\delta(n-1) + \delta(n-2) + 3\left(\delta(n-1) + 2\delta(n-2) + \delta(n-3)\right)$ $= \delta(n) + 5\delta(n-1) + 7\delta(n-2) + 3\delta(n-3)$ Peak value of output is 7
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0

`

Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	575
Type	MCQ
Body	The response of an initially relaxed linear network to a unit step function applied at $t=0$ is $s(t)=2\left[1-e^{-2t}\right]u(t)$. The response of this network to a unit impulse function will be
Option	$e^{-2t}u(t)$
Option	$2e^{-2t}u(t)$
Option	$3e^{-2t}u(t)$
Option	$4e^{-2t}u(t)$
Correct	4
Explanation	Impulse response can be obtained by differentiating the step response. $h(t)=\frac{d}{dt}2\left(1-e^{-2t}\right)u(t)$ $2\left(1-e^{-2t}\right)\delta(t)+2\left(0-e^{-2t}(-2)\right)u(t)$ $2\left(1-e^{-2(0)}\right)\delta(t)+4e^{-2t}u(t)$ $h(t)=4e^{-2u}u(t)$

`

Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	576
Type	MCQ
Body	A continuous-time system is described by $y(t)=e^{-\lvert x(t)\rvert}$, where $y(t)$ is the output and $x(t)$ is the input. The system is
Option	linear and time invariant
Option	linear and time variant
Option	non-linear and time invariant
Option	non-linear and time variant
Correct	3
Explanation	We know that $e^{a+b}=e^{a}\cdot e^{b}\neq e^{a}+e^{b}$ the above system does not obey additivity principle hence non-linear Output due to delayed input $(x(t-t_0))$ is $e^{-\lvert x(t-t_0)\rvert}$ _____(1)

`

	Delayed output response is $e^{-\|x(t-t_0)\|}$ _____(2) From (1) and (2) we can conclude that the system is time invariant system.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	577
Type	MCQ
Body	A linear time-invariant system has a frequency response, $H\left(e^{j\omega}\right)=e^{j\omega}\cdot\frac{1}{1+\cos\omega}$ then the casual LCCDE that connects the input to the output is
Option	$2y(n-1)+y(n)+2y(n+2)=x(n)$
Option	$2y(n)+y(n+1)+y(n-1)=2x(n+1)$
Option	$2y(n-1)+y(n)+y(n-2)=2x(n)$
Option	$2y(n+1)+y(n)+2y(n-2)=x(n-1)$

`

Correct	3
Explanation	Given, $H\left(e^{j\omega}\right)=\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}+\frac{1}{2}e^{-j\omega}}$ $\frac{Y\left(e^{j\omega}\right)}{X\left(e^{j\omega}\right)}=\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}+\frac{1}{2}e^{-j\omega}}$ $\Rightarrow Y\left(e^{j\omega}\right)+\frac{1}{2}e^{j\omega}Y\left(e^{j\omega}\right)+\frac{1}{2}e^{-y\omega}Y\left(e^{j\omega}\right)=e^{j\omega}X\left(e^{j\omega}\right)$ Taking inverse DTFT, $y(n)+\frac{1}{2}y(n+1)+\frac{1}{2}y(n-1)=x(n+1)$ ....(i) Put $n=n-1$ in equation (i), $2y(n-1)+y(n)+y(n-2)=2x(n)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	578
Type	MCQ

`

Body	If Laplace transform $F(s)=\frac{s+2}{s^2+8s+25}$, then the inverse Laplace transform $f(t)$ is
Option	$e^{-4t}\cos 3t-\frac{2}{3}e^{-4t}\sin 3t$
Option	$e^{-4t}\cos 3t+\frac{2}{3}e^{-4t}\sin 3t$
Option	$e^{-4t}\sin 3t-\frac{2}{3}e^{-4t}\cos 3t$
Option	$e^{4t}\cos 3t-\frac{2}{3}e^{-4t}\sin t$
Correct	1
Explanation	Given, $F(s)=\frac{s+2}{s^2+8s+25}=\frac{s+2}{(s+4)^2+(3)^2}$ $=\frac{s+4-2}{(s+4)^2+(3)^2}$ $=\frac{s+4}{(s+4)^2+(3)^2}-\frac{2}{(s+4)^2+(3)^2}$ By using inverse Laplace transform, $f(t)=e^{-4t}\cos 3t-\frac{2}{3}e^{-4t}\sin 3t$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-

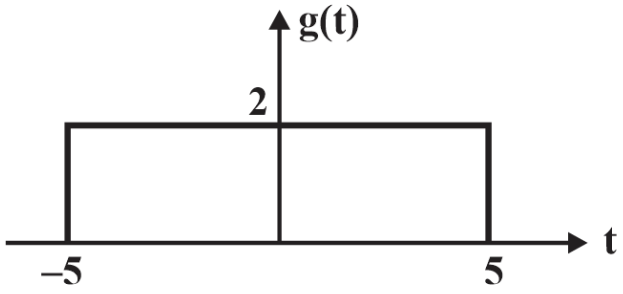
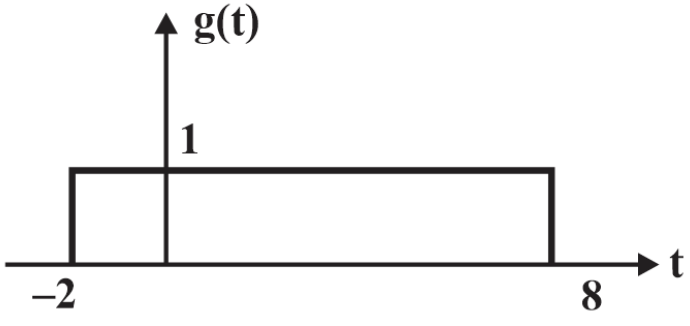
`

Video Solution	-
PYQ	
OTS ID	

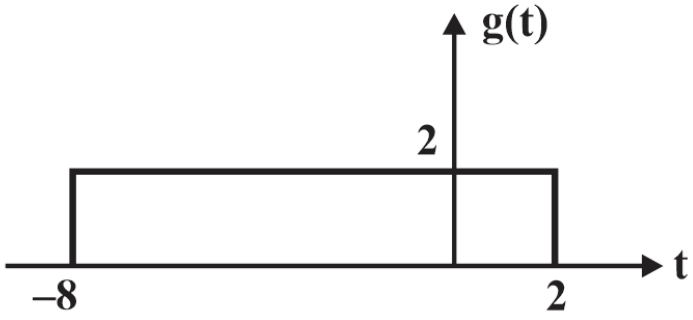
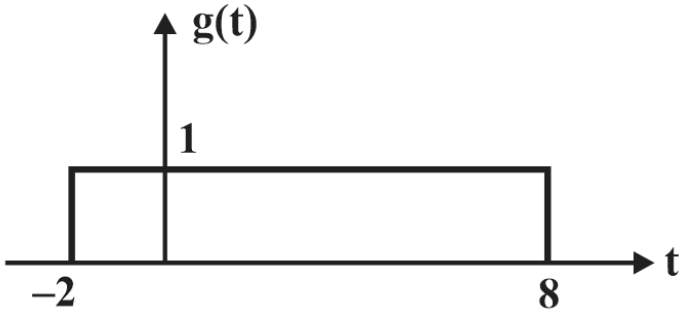
Question	579
Type	MCQ
Body	If $x(n)$ is real-valued periodic sequence with a period $N.x(n)$ and $X(k)$ form N -point Discrete Fourier Transform (DFT) pairs. Then the DFT of $y(k)$ of the sequence $y(n)=\frac{1}{N}\sum_{r=0}^{N-1}x(r)x(n+r)$ is
Option	0
Option	$\frac{1}{N}\sum_{r=0}^{N-1}X(r)X(k+r)$
Option	$\frac{1}{N}\sum_{r=0}^{N-1}X(k)X(k+r)$
Option	$\lvert X(k)\rvert^2$
Correct	4
Explanation	Given, $y(n)=\frac{1}{N}\sum_{r=0}^{N-1}x(r)x(n+r)$ It is auto correlation function, Hence, $y(n)=r_{XX}(n)\xleftrightarrow{DFT}\lvert X(k)\rvert^2$
Subject	Signals and Systems

`

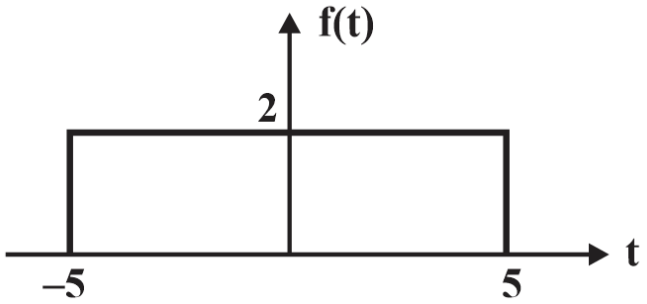
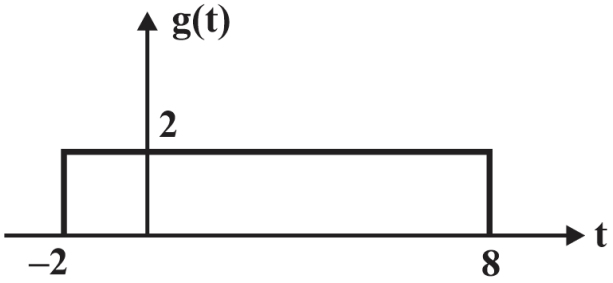
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	580
Type	MCQ
Body	The inverse Fourier transform of $G(\omega) = 20\frac{\sin 5\omega}{5\omega}e^{-j3\omega}$ is
Option	
Option	

`

Option	
Option	
Correct	4
Explanation	Given, $G(\omega) = 20\frac{\sin 5\omega}{5\omega}e^{-j3\omega}$ Let, $F^{-1}\left\{2a\frac{\sin a\omega}{a\omega}\right\} = f(t)$ Put $a = 5$, $F^{-1}\left\{10\frac{\sin 5\omega}{5\omega}\right\} = f(t)$ From linearity property, $f(t) = F^{-1}\left\{2\times 10\frac{\sin 5\omega}{5\omega}\right\} = 2f(t)$ $\therefore \quad F^{-1}\left\{20\frac{\sin 5\omega}{5\omega}\right\} \overset{FT}{\longleftrightarrow}$

`

	 $\therefore \quad g(t) = F^{-1}\left\{20\frac{\sin 5\omega}{5\omega}\cdot e^{-j3\omega}\right\} \xleftrightarrow{FT} 2f(t-3)$ $\therefore g(t) =$ 
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	581
Type	NAT

`

Body	Two sequences $x_1[n]$ and $x_2[n]$ have the same energy. Suppose $x_1[n] = \alpha 0.5^n u[n]$, where a is a positive real number and $u[n]$ is the unit step sequence. Assume, $x_2[n] = \begin{cases} \sqrt{1.2\alpha}, & \text{for } n = 0,1 \\ 0 & \text{otherwise} \end{cases}$ Then the value of a is ____ . (Answer upto two decimal places)
Correct	1.80 (1.75-1.85)
Explanation	Given, $x_1[n] = \alpha 0.5^n u[n]$ $x_2[n] = \begin{cases} \sqrt{1.2\alpha}, & \text{for } n = 0,1 \\ 0 & \text{otherwise} \end{cases}$ Energy of $x_1[n] = \sum_{n=-\infty}^{\infty} \lvert x_1[n] \rvert^2$ $= \sum_{n=-\infty}^{\infty} \alpha^2 \left(\frac{1}{4}\right)^n = \alpha^2 \cdot \frac{1}{1-\frac{1}{4}} = \alpha^2 \cdot \frac{4}{3}$ Energy of $x_2[n] = 1.2\alpha + 1.2\alpha = 2.4\alpha$ $\Rightarrow \alpha^2 \cdot \frac{4}{3} = 2.4\alpha$ $\alpha = \frac{3}{4} \times 2.4 = 1.8$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

Question	582
Type	NAT
Body	If $x[n]=\{1,2,3,2,1\}$ and $x[n]\xleftrightarrow{DTFT}X(e^{j\omega})$. Then the value of $\int_{-\pi}^{\pi}\left\|X(e^{j(\omega-\pi)})\right\|^2 d\omega$ is _____ .
Correct	119.38 (118.00-121.00)
Explanation	$\frac{1}{2\pi}\int_{-\pi}^{\pi}\left\|X(e^{j\omega})\right\|^2 d\omega=\sum_{n=-\infty}^{\infty}\|X[n]\|^2$ $\int_{-\pi}^{\pi}\|X(e^{j(\omega-\pi)}\|^2 d\omega=2\pi\sum_{n=-\infty}^{\infty}\|x(n)\|^2$ $\because x(n)e^{jn\pi}\xleftrightarrow{DTFT}X(e^{j(\omega-\pi)})$ $=2\pi[1+4+9+4+1]$ $=38\pi$ $=119.38$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

Question	583
Type	NAT
Body	A first-order discrete time system is described by the difference equation, $y[n]=\frac{1}{2}\left[x(n)+x(n-1)+y(n-1)\right]$ The impulse response at $n=0$ is _____ . (Answer upto one decimal place)
Correct	$0.5(0.3-0.7)$
Explanation	Given, $y[n]=\frac{1}{2}\left[x(n)+x(n-1)+y(n-1)\right]$ taking z-transform, $Y(z)=\frac{1}{2}\left[X(z)+z^{-1}X(z)+z^{-1}Y(z)\right]$ $H(z)=\frac{Y(z)}{X(z)}=\frac{0.5\left(1+z^{-1}\right)}{1-0.5z^{-1}}=\frac{(z+1)0.5}{z-0.5}$ $\frac{H(z)}{z}=\frac{(z+1)0.5}{z(z-0.5)}=\frac{A}{z}+\frac{B}{z-0.5}.$ $A=\frac{0.5(z+1)}{z-0.5}\Big\vert_{z=0}=-1$ $B=\frac{0.5(z+1)}{z}\Big\vert_{z=0.5}=-1.5$ $\therefore \quad H(z)=-1+\frac{1.5z}{z-0.5}$ $\therefore \quad h(n)=-\delta[n]+1.5(0.5)^n u[n]$

`

	at $n = 0$, $h(0) = -1 + 1.5 = 0.5$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

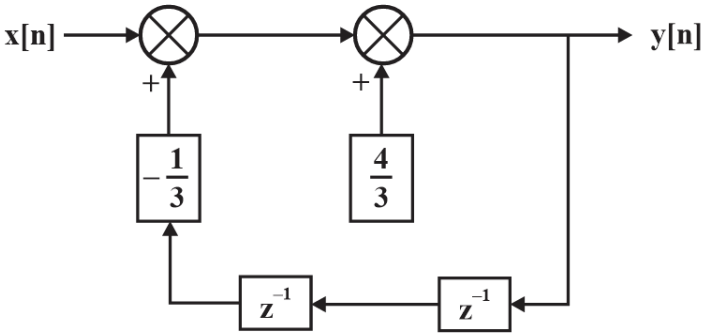
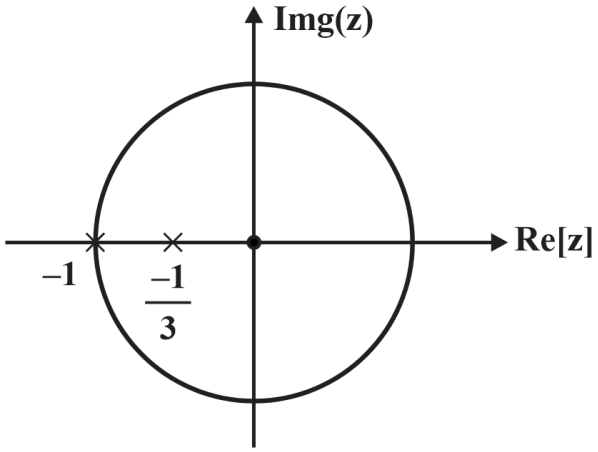
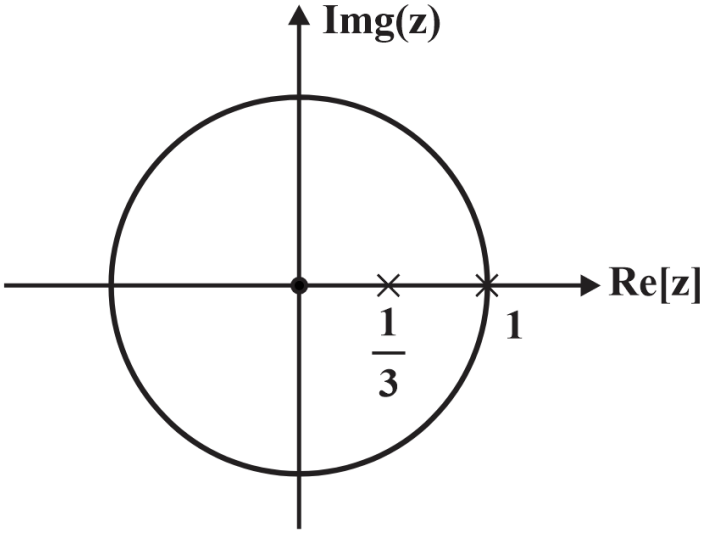
Question	584
Type	MSQ
Body	A discrete time signal $x(n) = \delta(n-1) + 2\delta(n-2)$ has z-transform $X(z)$. If $Y(z) = X\left(-\frac{z}{2}\right)$ is the z-transform of another signal $y[n]$, then
Option	$y(1) = 2$
Option	$y(-1) = 0$
Option	$y(2) = 8$
Option	$y(-2) = 8$
Correct	2,3

`

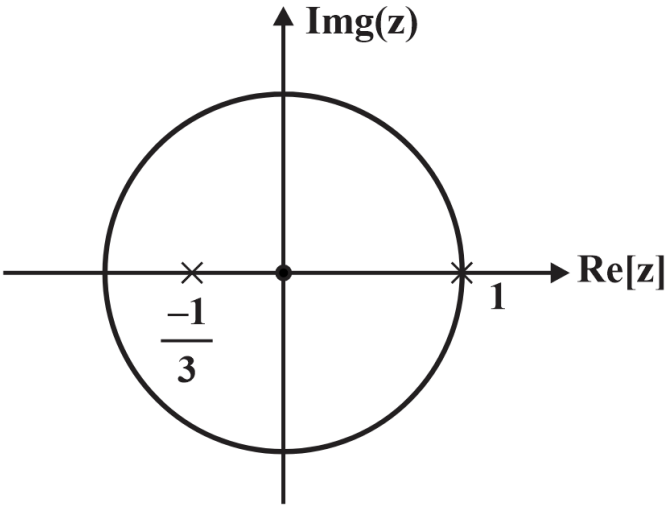
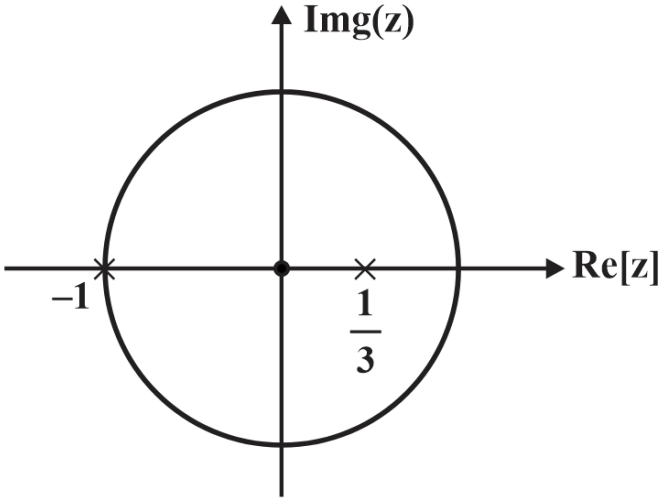
Explanation	Given, $x(n)=\delta(n-1)+2\delta(n-2)$ by taking z-transform, $X(z)=z^{-1}+2z^{-2}$ also given, $Y(z)=X\left(-\frac{z}{2}\right)$ $\therefore\ X\left(-\frac{z}{2}\right)=\left(-\frac{z}{2}\right)^{-1}+2\left(\frac{-z}{2}\right)^{-2}$ $=-2z^{-1}+8z^{-2}$ $\therefore\ Y(z)=-2\left(z^{-1}-4z^{-2}\right)$ $y(n)=-2\delta(n-1)+8\delta(n-2)$ $y(-1)=-2\delta(-2)+8\delta(-3)=0$ $y(2)=-2\delta(1)+8\delta(0)=8$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	585
Type	MCQ
Body	Consider the discrete-time system shown in the figure.

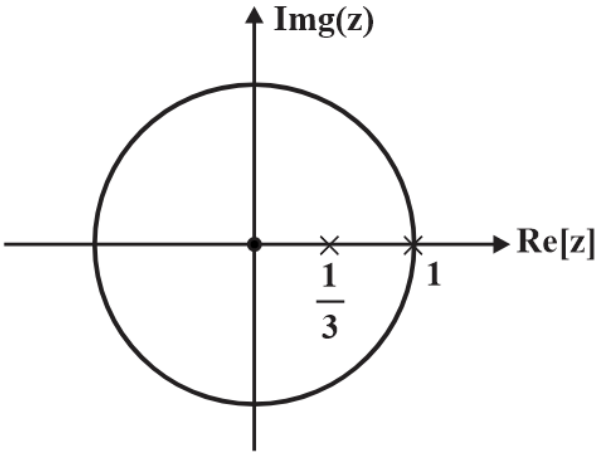
`

	 The pole-zero plot of the above system is
Option	
Option	

`

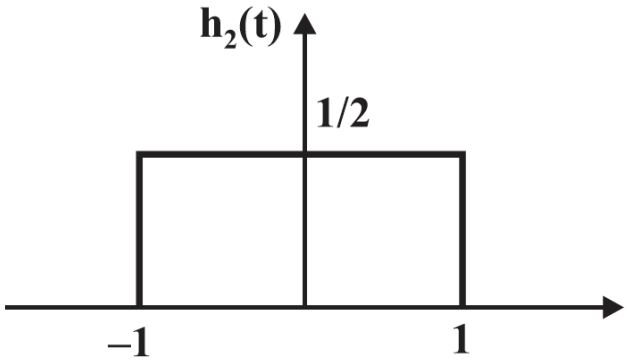
Option	
Option	
Correct	2
Explanation	The difference equation of the system $x(n)-\frac{1}{3}y(n-2)+\frac{4}{3}y(n-1)=y(n)$ by taking z -transform $X(z)-\frac{1}{3}z^{-2}Y(z)+\frac{4}{3}z^{-1}Y(z)=Y(z)$ $H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1+\frac{1}{3}z^{-2}-\frac{4}{3}z^{-1}}=\frac{1}{1+\frac{1}{3}\times\frac{1}{z^2}-\frac{4}{3}\frac{1}{z}}$

`

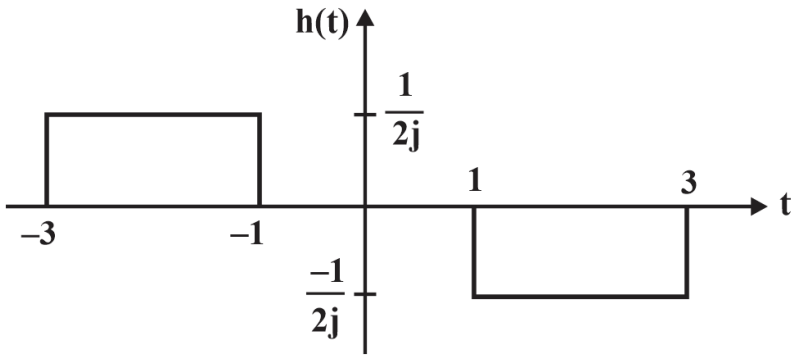
	$= \dfrac{z^2}{z^2 - \dfrac{4}{3}z + \dfrac{1}{3}}$ Poles are at, $z1, \dfrac{1}{3}$ 
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	586
Type	NAT

`

Body	The Fourier transform of a signal $h(t)$ is $H(\omega)=2\sin(2\omega)\cdot\frac{(\sin\omega)}{\omega}$, then the value of $\lvert h(2)\rvert$ is ____ .
Correct	$-0.5(-0.6--0.4)$
Explanation	Given, $H(\omega)=2\sin(2\omega)\cdot\frac{(\sin\omega)}{\omega}$ Let, $H_1(\omega)=2\sin(2\omega)$ $H_2(\omega)=\frac{\sin\omega}{\omega}$ From the property of Fourier transform, $A\,\text{rect}\left(\frac{t}{T}\right)\xleftrightarrow{FT}AT\sin c\left(\frac{\omega T}{2}\right)$ $AT\frac{\sin\left(\frac{\omega T}{2}\right)}{\left(\frac{\omega T}{2}\right)}$ on comparing, $\frac{T}{2}=1 \quad\Rightarrow T=2$ $AT=1\Rightarrow A=\frac{1}{T}=\frac{1}{2}$ $\therefore\quad H_2(\omega)=\frac{\sin\omega}{\omega}\xleftrightarrow{F.T.}$ 

`

	$h_1(t) = 2\sin(2\omega) = \dfrac{e^{+j2\omega} - e^{-j2\omega}}{j}$ $\therefore \quad H(\omega) = 2\sin(2\omega)H_2(\omega) = \dfrac{\left(e^{j2\omega} - e^{-j2\omega}\right)}{j}H_2(\omega)$ $H(\omega) = \left[e^{j2\omega}H_2(\omega) - e^{-j2\omega}H_2(\omega)\right]/j$ $h(t) = \left[h_2(t+2) - h_2(t-2)\right]\dfrac{1}{j}$  $\therefore \quad \lvert h(2)\rvert = 0.5$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	587

`

Type	NAT
Body	Consider the function $g(t)=e^{-2t}\cos(2\pi t)u(t)$ where $u(t)$ is the unit step function. Then the area under $g(t)$ is _____ (Answer upto three decimal places)
Correct	0.046 (0.035-0.055)
Explanation	Given, $g(t)=e^{-2t}\cos(2\pi t)$ $\text{Area}=\int_{-\infty}^{\infty}g(t)dt=\int_{-\infty}^{\infty}\cos(2\pi t)u(t)\cdot e^{-2t}dt$ $=G(s)\|_{s=0}$ $\therefore \quad G(s)=\frac{(s+2)}{(s+2)^2+(2\pi)^2}$ $\therefore \quad \text{Area}=\frac{s+2}{(s+2)^2+(4\pi)^2}\|_{s=0}=\frac{2}{4+4\pi^2}$ $\therefore \quad \text{Area}=0.046$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	588

`

Type	MSQ
Body	Consider the discrete time sequence given below: $x(n)=\{1,2,0,3,-2,4,7,5\}$ Then which of the following facts are correct regarding its DFT?
Option	$X(0)=20$
Option	$\sum_{K=0}^{7}X(k)=8$
Option	$X(4)=8$
Option	$\sum_{K=0}^{7}\|X(k)\|^2=764$
Correct	1,2
Explanation	(i) Given, $x(n)=\{1,2,0,3,-2,4,7,5\}$ From the definition of DFT, $X(k)=\sum_{n=0}^{N-1}x(n)e^{-j\frac{2\pi Kn}{N}}$ Given, $N=8$ $\frac{7}{2}\cdots$ $X(0)=\sum_{n=0}^{7}x(n)=1+2+0+3-2+4+7+5=20$ (ii) $X(4)=\sum_{n=0}^{7}x(n)\cdot e^{-j2\pi 4n/8}$ $=\sum_{n=0}^{7}x(n)\cdot e^{-j\pi n}=\sum_{n=0}^{7}x(n)(-1)^n$ $X(4)=1-2+0-3-2-4+7-5=-8$

`

	(iii) $\sum_{k=0}^{7} X(k)$ We know that inverse DFT is $x(n)=\frac{1}{N}\sum_{n=0}^{N-1} X(k)e^{j2\pi kn/N}$ Put $x=0, \quad N=8$ $x(0)=\frac{1}{N}\sum_{n=0}^{7} X(k)$ $\sum_{n=0}^{7} X(k)=Nx(0)=8\times 1=8$ (iv) From the expression for parreval's theorem is, $\sum_{n=0}^{N-1} \lvert x(n)\rvert^2=\frac{1}{N}\sum_{K=0}^{N-1}(X(k))^2$ Put $N=8,$ $\sum_{n=0}^{7}\left\lvert x(n)\right\rvert^2=\frac{1}{8}\sum_{k=0}^{7}\left\lvert X(k)\right\rvert^2$ $\sum_{k=0}^{7} \lvert X(k)\rvert^2=8\sum_{n=0}^{7} \lvert X(n)\rvert^2$ $=8[1+4+0+9+4+16+49+25]$ $=864$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	589
Type	NAT
Body	The discrete time Fourier transform of a signal $x(n)$ is $X(e^{j\omega}) = (1+\cos\omega)e^{-j\omega}$. Consider that $x_p(n)$ is a periodic signal of period $N = 5$ such that $x_p(n) = \begin{cases} x[n] & \text{for } n = 0,1,2 \\ 0 & \text{for } n = 3,4 \end{cases}$. Then the magnitude of the Fourier series coefficient a_3 is _____ . Answer upto 3 decimal places) (Note that $x_p(n) = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}Kn}$)
Correct	0.038 (0.030-0.045)
Explanation	Given, $x_p(n) = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}Kn}$ We know that, $a_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi}{N}Kn}$ $a_k = \frac{1}{N} X(e^{j\omega})\Big|_{\omega=\frac{2\pi k}{N}} = \frac{1}{N}(1+\cos\omega)e^{-j\omega}\Big|_{\omega=\frac{2\pi k}{N}}$ $= \frac{1}{N}\left(1+\cos\frac{2\pi\times 3}{N}\right)e^{\frac{j2\pi\times 3}{5}}$ $a_3 = \frac{1}{5}\left(1+\cos\frac{6\pi}{5}\right)e^{-j\frac{6\pi}{5}}$ $\lvert a_3\rvert = \frac{1}{5}\left(1+\cos\frac{6\pi}{5}\right) = 0.038$
Subject	Signals and Systems

`

Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	590
Type	MCQ
Body	The z-transform of $e^{-t}u(t)$ which is sampled at 10 Hz will be
Option	$\frac{z}{z-1}$
Option	$\frac{z}{z-0.1}$
Option	$\frac{z}{z-10}$
Option	$\frac{z}{z-0.9}$
Correct	4
Explanation	Given, $x(t)=e^{-t}u(t)$ Putting, $t=nT=n\times\frac{1}{10}=0.1n$

`

	$\therefore \quad x(n)=\begin{cases} e^{-0.1n} & \text{for } n \geq 0 \\ 0 & \text{for } n<0 \end{cases}$ $\therefore \quad x(n)=e^{-0.1n}u(n)$ $\therefore \quad X(z)=\frac{z}{z-e^{-0.1}}=\frac{z}{z-0.9}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

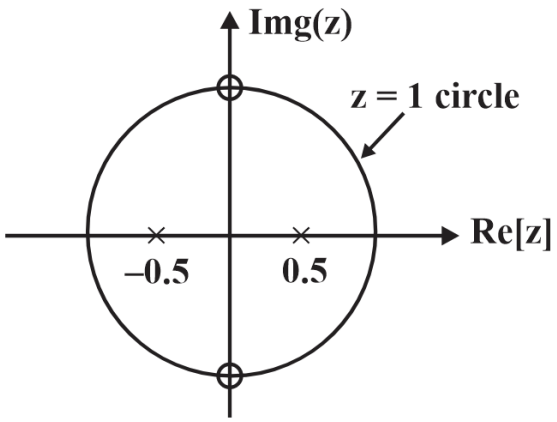
Question	591
Type	NAT
Body	The z-transform of a signal is given by $X(z)=\frac{1}{2}\frac{z^{-1}\left(1-z^{-4}\right)}{\left(1-z^{-1}\right)^2}$. , its final value is _____ .
Correct	2
Explanation	Given, $X(z)=\frac{1}{2}\frac{z^{-1}\left(1-z^{-4}\right)}{\left(1-z^{-1}\right)^2}$ Final value theorem,

`

	$x(\infty)=\lim_{z\to 1}\left(1-z^{-1}\right)X(z)$ $\therefore \quad x(\infty)=\lim_{z\to 1}\left(1-z^{-1}\right)\times\frac{1}{2}\frac{z^{-1}\left(1-z^{-4}\right)}{\left(1-z^{-1}\right)^2}$ $=\lim_{z\to 1}\frac{1}{2}\frac{z^{-1}\left(1-z^{-4}\right)}{\left(1-z^{-1}\right)}$ $=\frac{1}{2}\times\lim_{z\to 1}\frac{\left(z^2-1\right)\left(z^2+1\right)}{z^4\left(z-1\right)}$ $=\frac{1}{2}\times\lim_{z\to 1}\frac{\left(z+1\right)\left(z^2+1\right)}{z^4}$ $=\frac{1}{2}\times 2\times 2=2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	592
Type	MSQ

`

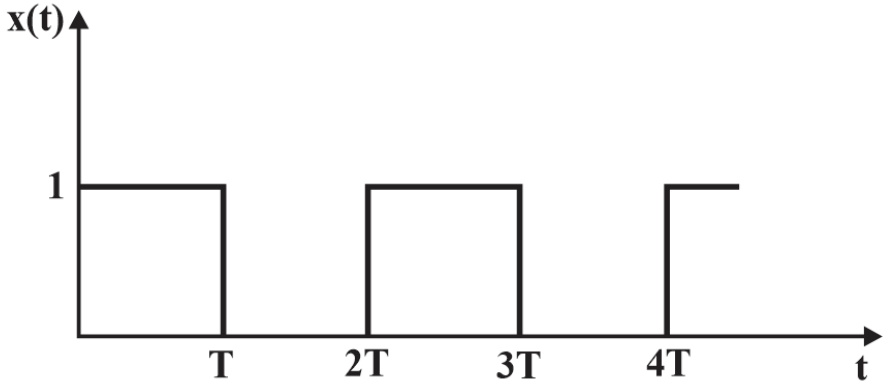
Body	Consider the linear discrete time system, $H(z)=\frac{(z^2+1)}{[(z+0.5)(z-0.5)]}$. Which of the following facts are correct?
Option	The system is stable
Option	$h(0)=4$
Option	$h(\infty)=0$
Option	$h(0)=-4$
Correct	1,3
Explanation	Given, $H(z)=\frac{(z^2+1)}{[(z+0.5)(z-0.5)]}$ Poles, $z=0.5$ and -0.5 Zeros, $z=\pm j1$ Pole-zero plot,  Since all poles are lying inside the unit circle $(z=1)$. Therefore, the system is stable For the initial value theorem,

`

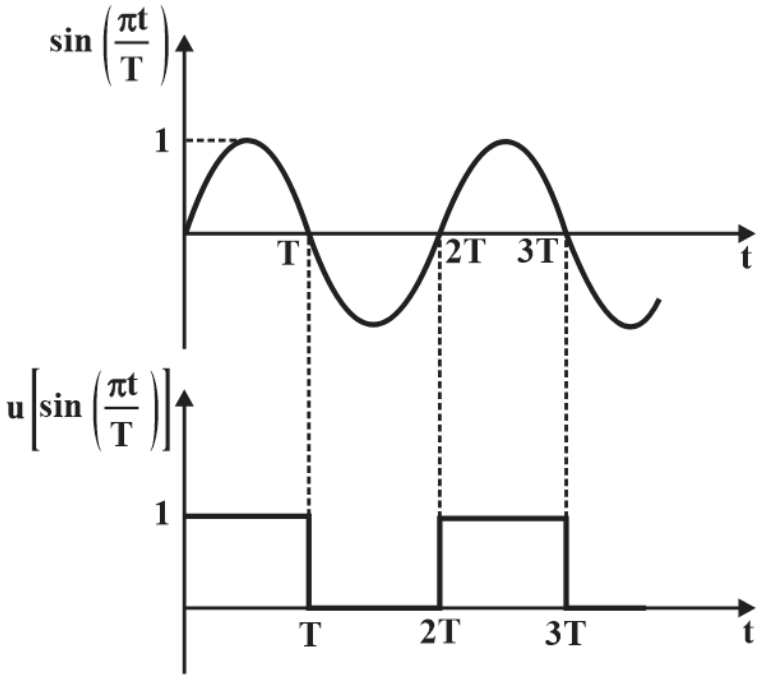
	$h(0)=\lim_{z\to\infty}H(z)=\lim_{z\to\infty}\frac{z^2+1}{(z+0.5)(z-0.5)}=1$ From the final value theorem, $h(\infty)=\lim_{z\to\infty}(z-1)H(z)$ $=\lim_{z\to\infty}(z-1)\frac{z^2+1}{(z+0.5)(z-0.5)}=0$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	593
Type	MCQ
Body	A continuous-time periodic signal is shown below in figure.

`

	 The signal can be described by the function
Option	$\operatorname{sgn}\left(\sin\frac{\pi t}{T}\right)$
Option	$\sin\left[\operatorname{sgn}(t)\right]$
Option	$u\left[\sin\left(\frac{\pi t}{T}\right)\right]$
Option	$\sin\left(u(t)\right)$
Correct	3
Explanation	$u(t)=\begin{cases}1; & t\geq 0\\ 0; & t<0\end{cases}$ $\therefore \quad u\left[\sin\left(\frac{\pi t}{T}\right)\right]=\begin{cases}1; & \sin\left(\frac{\pi t}{T}\right)\geq 0\\ 0; & \sin\left(\frac{\pi t}{T}\right)<0\end{cases}$

`

	
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	594
Type	MCQ

`

Body	The z-transform of signal $x[n]=e^{jn\frac{\pi}{2}}u[n]$ is
Option	$\frac{z}{z-j};ROC:\lvert z\rvert>1$
Option	$\frac{z}{z+j};ROC:\lvert z\rvert>1$
Option	$\frac{z}{z+1};ROC:\lvert z\rvert>1$
Option	$\frac{z}{z-1};ROC:\lvert z\rvert>1$
Correct	1
Explanation	$X(z)=\sum_{n=-\infty}^{\infty}e^{jn\frac{\pi}{2}}z^{-n}u(n)=\sum_{n=0}^{\infty}\left(e^{j\frac{\pi}{2}}z^{-1}\right)^n$ $=\frac{1}{1-e^{j\frac{\pi}{2}}z^{-1}};\lvert z\rvert>\left\lvert e^{j\frac{\pi}{2}}\right\rvert=\frac{z}{z-j};ROC:\lvert z\rvert>1$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	595
Type	MCQ
Body	The trigonometric Fourier series coefficient a_n in terms of exponential Fourier series coefficient C_n is
Option	$a_n = C_n + C_{-n}$
Option	$a_n = C_n - C_{-n}$
Option	$a_n = C_n + jC_{-n}$
Option	$a_n = C_n - jC_{-n}$
Correct	1
Explanation	$C_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}dt$ $C_{-n} = \frac{1}{T}\int_0^T x(t)e^{jn\omega_0 t}dt = C_n^*$ $C_n = \frac{a_n - jb_n}{2}$ $C_{-n} = C_n^* = \frac{a_n + jb_n}{2}$ $a_n = C_n + C_{-n}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	

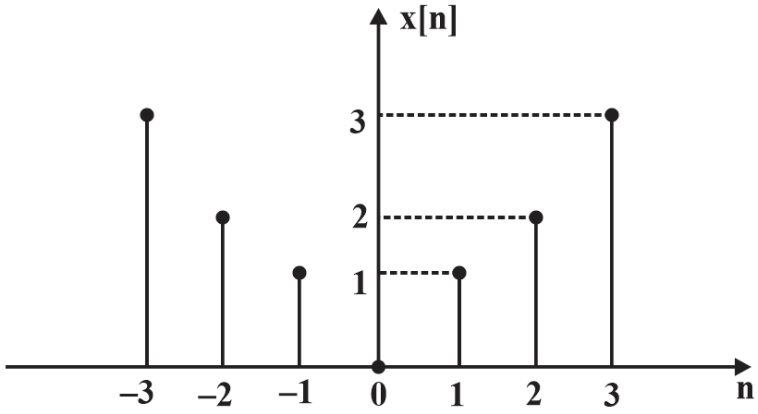
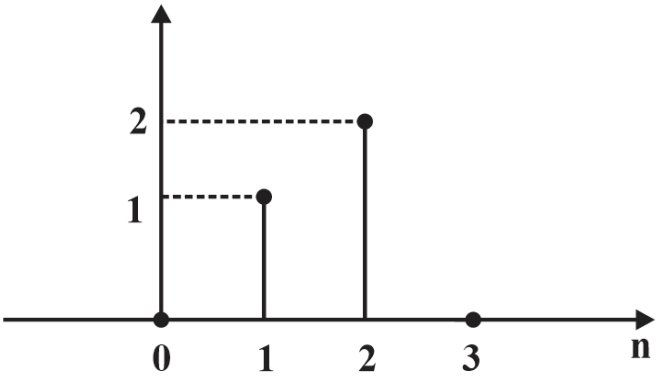
`

OTS ID	

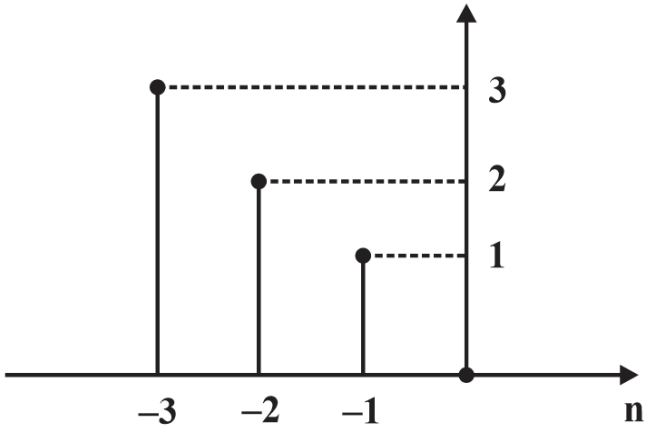
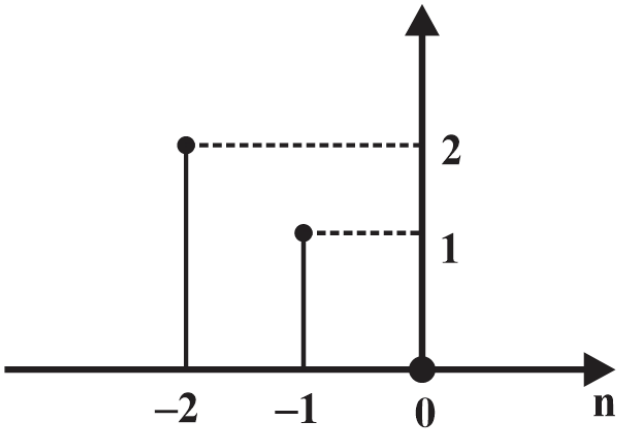
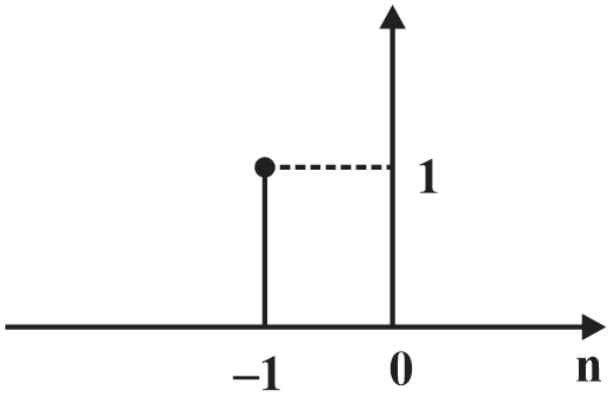
Question	596
Type	MCQ
Body	The time domain signal with time period, $T = 2$ whose Fourier series coefficient is given by $C_n = j\delta(n-1) - j\delta(n+1) + \delta(n-3) + \delta(n+3)$ will be
Option	$2\sin 2\pi t - \sin 3\pi t$
Option	$-2\sin \pi t + 2\cos 3\pi t$
Option	$\sin 2\pi t + 2\sin 3\pi t$
Option	$\cos \pi t - 2\sin 3\pi t$
Correct	2
Explanation	$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$ $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\pi t} \left\{ \because \omega_0 = \frac{2\pi}{T} = \pi \right\}$ $x(t) = \sum_{n=-\infty}^{\infty} \left[j\delta(n-1) - j\delta(n+1) + \delta(n-3) + \delta(n+3) \right] e^{jn\pi t}$ $x(t) = je^{\pi t} - je^{-\pi t} + e^{j3\pi t} + e^{-j3\pi t}$ $x(t) = -2\sin \pi t + 2\cos 3\pi t$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66

`

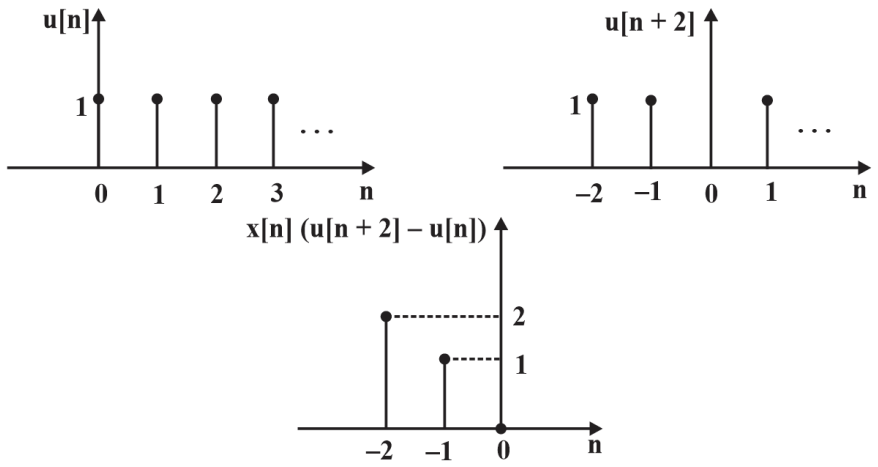
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	597
Type	MCQ
Body	A discrete-time signal $x[n]$ is shown below :  Which of the following signals represent $x[n](u[n+2]-u[n])$?
Option	

`

Option	
Option	
Option	
Correct	3

`

Explanation	
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	598
Type	NAT
Body	The impulse response of a discrete-time LTI system is given by $h[n]=\left(\frac{1}{2}\right)^n u[n]$. If $y[n]$ is the output of the system with the input $x[n]=2\delta[n]+\delta[n-3]$ then y[5] will be _____ (upto three decimal places)

`

Correct	0.3125 (0.300-0.320)
Explanation	$y[n] = x[n] * h[n]$ $= \{2\delta[n] + \delta[n-3]\} * \left(\frac{1}{2}\right)^n u(n)$ $= 2\delta[n] * \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^n u(n) * \delta[n-3]$ $= 2\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-3} u(n-3)$ $y[5] = 2\left(\frac{1}{2}\right)^5 u(5) + \left(\frac{1}{2}\right)^2 u(2) = \frac{1}{16} + \frac{1}{4} = \frac{5}{16}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

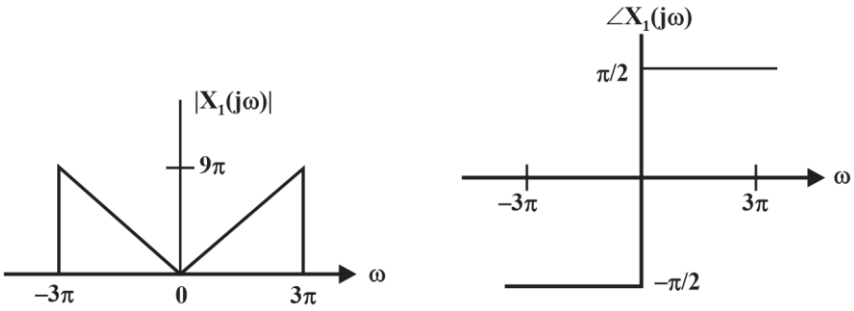
Question	599
Type	MCQ
Body	Let $x[n]$ be a sequence with z-transform $X(z)$. Then expression the $x_1(n) = x(2n)$ in terms of $X(z)$ is

`

Option	$\frac{1}{2}\left[X\left(\sqrt{z}\right)+X\left(\frac{-1}{\sqrt{z}}\right)\right]$
Option	$\frac{1}{2}\left[X\left(\sqrt{z}\right)+X\left(\frac{1}{\sqrt{z}}\right)\right]$
Option	$\frac{1}{2}\left[X\left(\sqrt{z}\right)-X\left(-\sqrt{z}\right)\right]$
Option	$\frac{1}{2}\left[X\left(\sqrt{z}\right)+X\left(-\sqrt{z}\right)\right]$
Correct	4
Explanation	Given, $x_1(n)=x(2n)$ From the definition of z -transform, $X(z)=\sum_{n=-\infty}^{\infty}x(n)z^{-n}$ $\therefore \quad X_1(z)=\sum_{n=-\infty}^{\infty}x(2n)z^{-n}$ $K=2n \quad \Rightarrow \quad n=\frac{K}{2}$ at $n=-\infty \quad \Rightarrow \quad K=-\infty$ $n=+\infty \quad \Rightarrow \quad K=+\infty$ $=\sum_{K=-\infty}^{\infty}x(K)z^{\frac{K}{2}}$ $=\sum_{K=-\infty}^{\infty}\left[\frac{x(K)+(-1)^K x(K)}{2}\right]z^{\frac{K}{2}}; K \text{ even}$ $=\frac{1}{2}\sum_{K=-\infty}^{\infty}x(K)z^{\frac{K}{2}}+\frac{1}{2}\sum_{K=-\infty}^{\infty}x(K)\left(-z^{\frac{1}{2}}\right)^{-K}$ From the definition of z -transform $X_1(z)=\frac{1}{2}\left[X\left(\sqrt{z}\right)+X\left(-\sqrt{z}\right)\right]$

`

Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

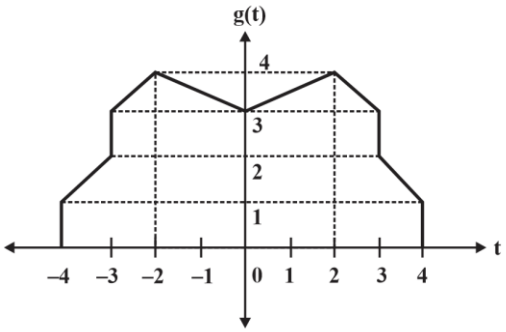
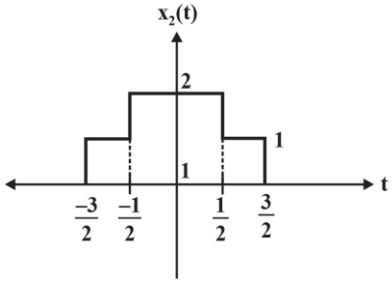
Question	600
Type	MCQ
Body	A signal $x_1(t)$ whose fourier transform is $X_1(j\omega)$ has the following magnitude and phase response as shown below:  Then the signal $x(t)$ is
Option	$\frac{3}{\pi t}(\cos 3\pi t - \sin 3\pi t)$
Option	$\frac{3}{\pi t}(\pi t\cos 3\pi t - \sin 3\pi t)$

`

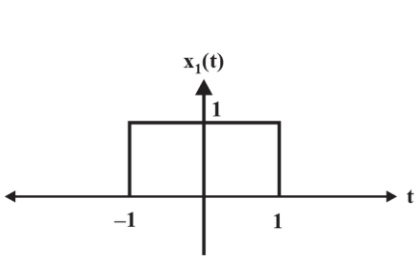
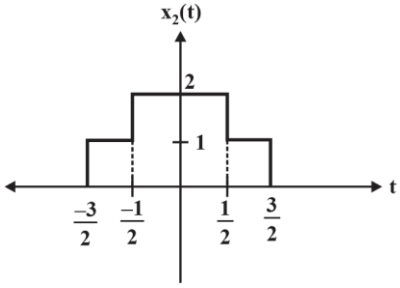
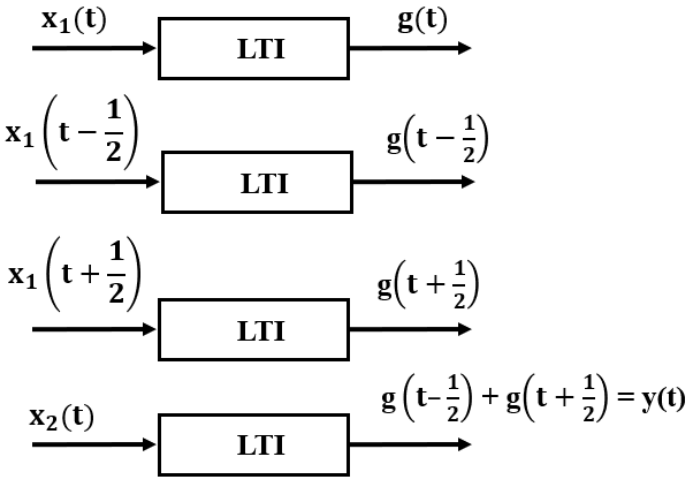
Option	$\frac{3}{\pi t^2}\left(3\pi t\cos 3\pi t-\sin 3\pi t\right)$
Option	$\frac{3}{\pi^2 t^2}\left(3t\cos 3\pi t-\sin 3\pi t\right)$
Correct	3
Explanation	The fourier transform of $x(t)$ can be written $X_1(j\omega)=\left\|X_1(j\omega)\right\|e^{j\angle X_1(j\omega)}$ Let, $X_{1a}(j\omega)=\begin{cases}1; & \left\|\omega\right\|<3\pi\\ 0; & \text{otherwise}\end{cases}$ Note that, given $X_1(j\omega)$ is " $3j\omega$ " times $X_{1a}(j\omega)$ $\therefore \quad X_1(j\omega)=\begin{cases}3j\omega;\left\|\omega\right\|<3\pi\\ 0;\text{otherwise}\end{cases}$ Since, at $\omega=3\pi,\left\|X_1(j\omega)\right\|=9\pi$ and $\angle X_1(j\omega)=\frac{\pi}{2}$ $\omega=-3\pi,\left\|X_1(j\omega)\right\|=9\pi$ and $\angle X_1(j\omega)=-\frac{\pi}{2}$ By taking inverse fourier transform, Thus, $x_1(t)=3\frac{d}{dt}x_{1a}(t)$ [By using differential property] also we can express $x_{1a}(t)=\frac{\sin 3\pi t}{\pi t}$ Thus, $x_1(t)=3\frac{d}{dt}\left[\frac{\sin 3\pi t}{\pi t}\right]$ $=\frac{3}{\pi}\times\frac{1}{t^2}\left[3\pi t\cos 3\pi t-\sin 3\pi t\right]$ $\therefore \quad x_1(t)=\frac{3}{\pi t^2}\left[3\pi t\cos 3\pi t-\sin 3\pi t\right]$
Subject	Signals and Systems

`

Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

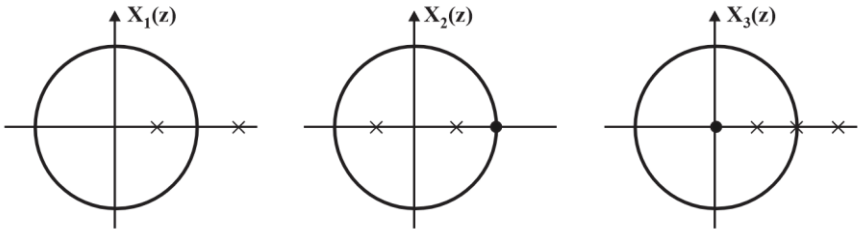
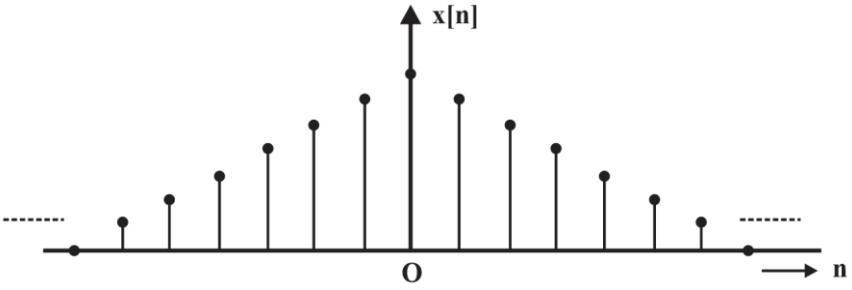
Question	601
Type	MCQ
Body	The response of an LTI system for input $x_1(t) = u(t+1) - u(t-1)$ is given as $g(t)$. If the input to the same system is $x_2(t)$ as shown below, and response being $y(t)$, then $y(0) =$ _____ .  
Option	6.5
Option	2
Option	3
Option	4

`

Correct	1
Explanation	  $x_2(t) = x_1\left(t-\frac{1}{2}\right) + x_1\left(t+\frac{1}{2}\right)$ Since the system is LTI  $y(0) = g\left(-\frac{1}{2}\right) + g\left(\frac{1}{2}\right) = 3.25 + 3.25 = 6.50$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

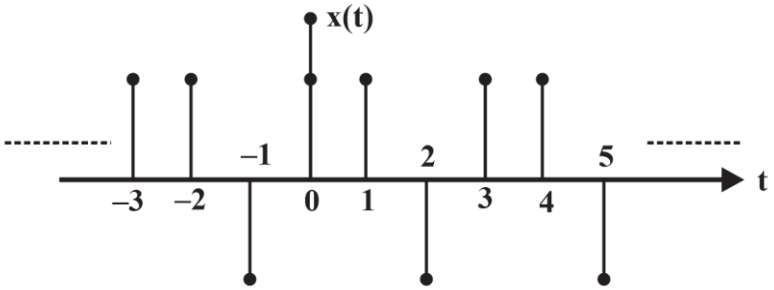
Question	602
Type	MCQ
Body	The following diagrams represent systems with poles (indicated by X s) and zero's (indicated by • 's). The scale for each diagram is indicated by the circle, which has unit radius as shown below  which of the above pole-zero diagram represents the system with the following unit-sample response 
Option	$X_1(z)$
Option	$X_2(z)$
Option	$X_3(z)$
Option	None of the given
Correct	4

`

Explanation	We can rewrite the given $x(n)$ signal $x(n)=a^{\|n\|}, 0<a<1$ The z-transform of the $x(n)$ is $X(z)$ $X(z)=\sum_{n=0}^{\infty}a^{n}z^{-n}+\sum_{n=-\infty}^{-1}a^{-n}z^{-n}$ (or) $x[n]=a^{n}u(n)+a^{-n}u[-n-1]$ $X(z)=\frac{1}{1-az^{-1}}-\frac{1}{1-a^{-1}z^{-1}};a<\|z\|<\frac{1}{a}$ (or) $\left(\because a^{-n}u[-n-1]\xleftrightarrow{Z}\frac{-1}{1-a^{-1}z^{-1}};\|z\|<\frac{1}{a}\right)$ $=\frac{z}{z-a}-\frac{z}{z-\frac{1}{a}};a<\|z\|<\frac{1}{a}=\frac{z^{2}-\frac{z}{a}-z^{2}+az}{(z-a)\left(z-\frac{1}{a}\right)}$ $X(z)=\frac{\left[a-\frac{1}{a}\right]z}{(z-a)\left(z-\frac{1}{a}\right)};a<\|z\|<\frac{1}{a}$ This z-transform has poles at $z=a,z=\frac{1}{a}$, and a zero at $z=0$. ∴ None of the given pole-zero diagram represents $x(n)$. Hence option (d) is correct.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-

`

Video Solution	-
PYQ	
OTS ID	

Question	603
Type	NAT
Body	Let the following periodic signal $x(t)=\sum_{m=-\infty}^{\infty}\delta(t-3m)+\delta(t-1-3m)-\delta(t-2-3m)$ be the input to an LTI system with system function $H(s)=e^{s/4}-e^{-s/4}$ Let b_k represents the fourier series coefficient of the resulting output signal $y(t)$. Then $\lvert b_3\rvert$ will be ____ . Answer upto three decimal places)
Correct	0.66 (0.65-0.67)
Explanation	Given, periodic signal $x(t)=\sum_{m=-\infty}^{\infty}\delta(t-3m)+\delta(t-1-3m)-\delta(t-2-3m)$ redrawing the periodic signal 

`

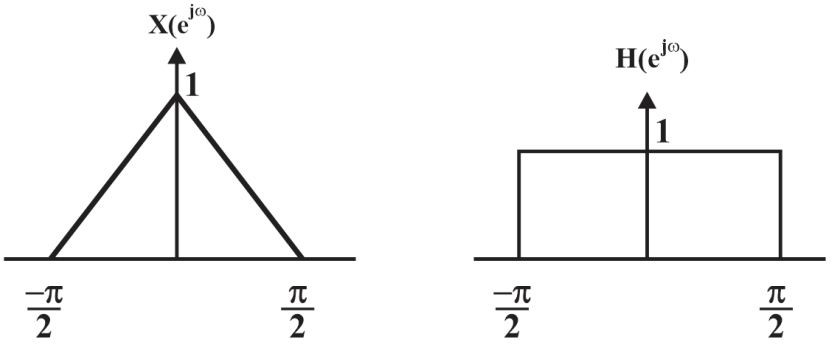
	Clearly the period of $x(t)$ is $T = 3$ $\therefore$ $y(t)$ is also periodic with period $T = 3$. The fundamental frequency of $x(t)$ is, $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{3}$ Let a_k represents the fourier series coefficient of $x(t)$. Then $a_k = \frac{1}{T}\int_0^T x(t)e^{-j\omega_0 kt}dt$ $= \frac{1}{3}\int_0^3 (\delta(t) + \delta(t-1) - \delta(t-2)e^{-j\frac{2\pi}{T}kt}dt$ $a_k = \frac{1}{3}\left[1 + e^{-j\frac{2\pi}{3}k} - e^{-j\frac{2\pi}{3}k\times 2}\right]$ at $k = 3$; $a_3 = \frac{1}{3}\left(1 + e^{-j2\pi} - e^{-j4\pi}\right)$ $a_3 = \frac{1}{3}$ The frequency response of the system is given, $H(j\omega) = e^{j\omega/4} - e^{-j\omega/4}$ $= 2j\sin\frac{\omega}{4}$ $\therefore$ $b_k = H\left(j\frac{2\pi}{3}k\right)a_k = \left(j2\sin\frac{2\pi}{12}k\right)a_k$ at $k = 3$ $b_3 = \left(j2\sin\frac{\pi}{2}\right)\frac{1}{3} = j\frac{2}{3}$ $\therefore$ $\lvert b_3\rvert = 0.66$
Subject	Signals and Systems
Topic	
Correct Marks	2

`

Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	604
Type	NAT
Body	Let the signal $x[n]$ be a discrete time signal represented as $x[n]=\left(\frac{\sin(\pi/4)n}{\pi n}\right)^2$ This signal is passed through a system whose impulse response is represented as $h[n]=\frac{\sin\omega_c n}{\pi n}$. The minimum value of $f_c\left(f_c=\frac{\omega_c}{2\pi}\right)$ so that $y[n]=x[n]$ is _____ Hz.
Correct	0.25
Explanation	$y[n]=x[n]*h[n]$ $=\left(\frac{\sin\frac{\pi}{4}n}{\pi n}\right)^2*\left(\frac{\sin\omega_c n}{\pi n}\right)$ Convolution in time domain specifies multiplication in frequency domain thus, $x[n]$ has a Fourier transform equal

`

	to a triangular pulse with width $\frac{\pi}{2}$ and the width of the filter $H\left(e^{j\omega}\right)$ should also have a limit of $\frac{\pi}{2}$ to have $x[n]=y[n]$.  $\therefore\ \omega_c=\frac{\pi}{2}$ $2\pi f_c=\frac{\pi}{2}$ $f_c=\frac{1}{4}=0.25\,\text{Hz}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

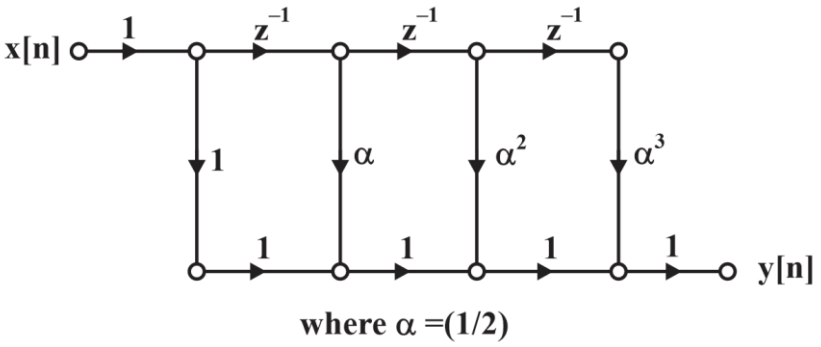
Question	605
Type	NAT
Body	A periodic sequence with period $N=4$ is defined as $x[n]=[0,1,2,3]$ for one period. It has discrete time Fourier series coefficient C_k. The value of C_2 is _____ (Answer upto two decimal places)
Correct	$-0.50(-0.55--0.45)$
Explanation	$C_k=\frac{1}{N}\sum_{n=0}^{N-1}x[n]e^{-jk\omega_0 n}$ $C_2=\frac{1}{4}\sum_{n=0}^{3}x[n]e^{-j2\times\frac{2\pi}{4}\times n}$ $=\frac{1}{4}\sum_{n=0}^{3}x[n](-1)^n=\frac{1}{4}[0+(-1)+2-3]$ $=-\frac{1}{2}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	606
Type	NAT
Body	If $x[k]=(4-k)$, then the value of $\sum_{k=1}^{3} x[k]\delta[1-k]$ is
Correct	3 (3.00-3.00)
Explanation	Let, $y=\sum_{k=1}^{3} x[k]\delta[1-k]$ $=\sum_{k=1}^{3}(4-k)\delta[1-k]$ $=\sum_{k=1}^{3}3\times\delta[1-k]$ ...from multiplication property of impulse $=3\times1=3$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	607
Type	NAT
Body	The direct form of an FIR filter is shown below:

`

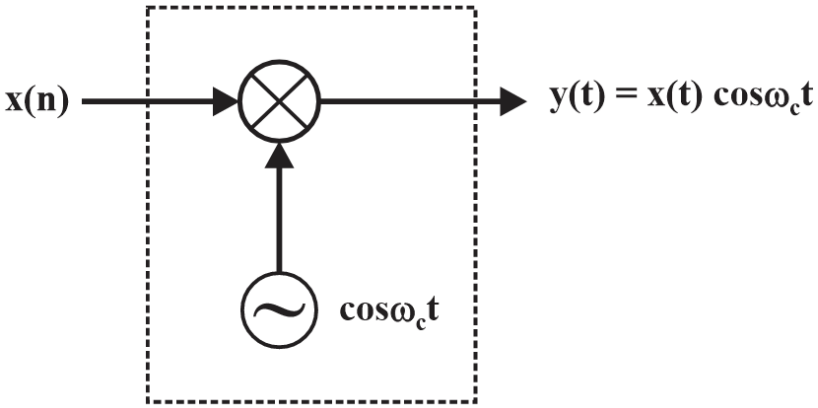
	 where α =(1/2) The input given is $x[n]=\delta[n]$ and the output is $y[n]$. Then the value of $y[2]$ will be _____ . (Answer upto two decimal places)
Correct	0.25 (0.24-0.26)
Explanation	If $x[n]=\delta[n]$, then $y[n]=h[n]$ So, $y[n]=h[n]=\delta[n]+\alpha\delta[n-1]+\alpha^2\delta[n-2]+\alpha^3\delta[n-3]$ $h[n]=\begin{cases}\alpha^n; & 0\le n\le 3\\ 0; & \text{otherwise}\end{cases}$ $\therefore\ y[2]=\alpha^2=\frac{1}{4}=0.25\left(\because \text{ Given that, } \alpha=\frac{1}{2}\right)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	608
Type	MCQ
Body	Consider the system defined by the difference equation $y[n]=ay[n-1]+bx[n]+x[n-1]$, where a and b are real and $\|a\|<1$. Then the relation between a and b that must exist if the frequency response is to have a constant magnitude for all 'ω ' i.e, $\left\|H\left(e^{j\omega}\right)\right\|=1$ is
Option	$b=-a$
Option	$b=a$
Option	$b=\frac{-a}{2}$
Option	$b=\frac{a}{2}$
Correct	1
Explanation	Given difference equation, $y[n]=ay[n-1]+bx[n]+x[n-1]$ By taking DTFT, $Y\left(e^{j\omega}\right)=ae^{-j\omega}Y\left(e^{j\omega}\right)+bX\left(e^{j\omega}\right)+e^{-j\omega}X\left(e^{j\omega}\right)$ $Y\left(e^{j\omega}\right)-ae^{-j\omega}Y\left(e^{j\omega}\right)=bX\left(e^{j\omega}\right)+e^{-j\omega}X\left(e^{j\omega}\right)$ $Y\left(e^{j\omega}\right)\left[1-ae^{-j\omega}\right]=\left(b+e^{-j\omega}\right)X\left(e^{j\omega}\right)$ $H\left(e^{j\omega}\right)=\frac{Y\left(e^{j\omega}\right)}{X\left(e^{j\omega}\right)}=\frac{b+e^{-j\omega}}{1-ae^{-j\omega}}=\frac{b+\cos\omega-j\sin\omega}{1-a\left(\cos\omega-j\sin\omega\right)}$

`

	$\left\|H\left(e^{j\omega}\right)\right\|^2 = \frac{1+b^2+2b\cos\omega}{1+a^2-2a\cos\omega}$ To satisfy $\left\|H\left(e^{j\omega}\right)\right\|^2 = 1$ only when $b = -a$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	609
Type	MSQ
Body	Consider the system shown in figure below :  The system is
Option	Memoryless

`

Option	Causal
Option	Linear
Option	Time-invariant
Correct	1,2,3
Explanation	$y(t) = T[x(t)] = x(t)\cos\omega_c t$ • Since the output $y(t)$ depends only on the present values of input $x(t)$, the system is memoryless. • Since $y(t)$ does not depend on the future values of input $x(t)$, the system is causal. • Let $x(t) = ax_1(t) + bx_2(t)$, then $y(t) = T\{x(t)\} = \{ax_1(t) + bx_2(t)\}\cos\omega_c t$ $= ax_1(t)\cos\omega_c t + bx_2(t)\cos\omega_c t$ $= ay_1(t) + by_2(t)$ ∴ The system is linear. • Let $y_1(t)$ be the output produced by shifted input $x_1(t) = x(t - t_o)$, then, $y_1(t) = T[x(t - t_o)] = x(t - t_o)\cos\omega_c t$ But $y(t - t_o) = x(t - t_o)\cos\omega_c(t - t_o) \neq y_1(t)$ ∴ The system is not time-invariant.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

Question	610
Type	MCQ
Body	The inverse Fourier transform of the signal $X(\omega) = e^{-2\|\omega\|}$ is
Option	$\frac{2}{\pi(4+t^2)}$
Option	$\frac{2}{2\pi(4+t^2)}$
Option	$\frac{2}{\pi(4+t^2)}$
Option	$\frac{1}{(4+t^2)}$
Correct	3
Explanation	Given, $X(\omega) = e^{-2\|\omega\|}$ We know that, $e^{-a\|t\|} \xleftrightarrow{F.T.} \frac{2a}{a^2+\omega^2}$ Put $a = 2$, $e^{-2\|t\|} \xleftrightarrow{F.T.} \frac{4}{4+\omega^2}$ By using duality property,

`

	$\frac{4}{4+t^2} \xleftrightarrow{\text{IFT}} 2\pi e^{-2\lvert-\alpha\rvert}$ $\frac{4}{2\pi\left(4+t^2\right)} \xleftrightarrow{\text{IFT}} e^{-2\lvert\alpha\rvert}$ $\therefore \quad \frac{2}{\pi\left(4+t^2\right)} \xleftrightarrow{\text{IFT}} e^{-2\lvert\alpha\rvert}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	611
Type	MCQ
Body	Let $x(t)=e^{-t}u(t)$. The inverse Laplace transform of $e^{-3s}X(2s)$ is equal to
Option	$\frac{1}{2}e^{-(t-3)/2}u(t+3)$
Option	$e^{-(t-3)/2}u(t-3)$

`

Option	$\frac{1}{2}e^{-(t+3)/2}u(t+3)$
Option	$\frac{1}{2}e^{-(t-3)/2}u(t-3)$
Correct	4
Explanation	Given, $x(t)=e^{-t}u(t)$ We know that, $x(at)\leftrightarrow\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$ Put $a=\frac{1}{2}$; $x\left(\frac{t}{2}\right)\leftrightarrow 2X(2s)$ $\therefore \quad X(2s)\leftrightarrow\frac{1}{2}x\left(\frac{t}{2}\right)=\frac{1}{2}e^{-t/2}u(t)$ From time shifting property, $x(t-t_0)\leftrightarrow e^{-st_0}X(s)$ Put $t_0=3$. $x(t-3)\leftrightarrow e^{-3s}X(s)$ $\therefore \quad L^{-1}\left[e^{-3s}X(2s)\right]=\frac{1}{2}e^{-(t-3)/2}u(t-3)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

Question	612
Type	MCQ
Body	If the region of z-transform of signal $x_1[n]+x_2[n]$ is $\frac{1}{4}<\|z\|<\frac{3}{4}$, then the region of convergence of z-transform of signal $x_1[n]-x_2[n]$ includes
Option	$\frac{1}{4}<\|z\|<\frac{3}{4}$
Option	$\frac{3}{4}<\|z\|<4$
Option	$3<\|z\|<4$
Option	$\frac{1}{4}<\|z\|<4$
Correct	1
Explanation	ROC remains unaffected for addition and subtraction operation in z-transform, i.e., $\rightarrow$ If ROC of $x_1[n]+x_2[n]$ is R then ROC of $x_1[n]-x_2[n]$ is also R.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66

`

Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	613
Type	NAT
Body	A signal $x(t)$ has the Fourier transform $X(j\omega)$ having the following facts : $F^{-1}\{(1+j\omega)X(j\omega)\} = Ae^{-2t}u(t)$ (where A is a real and positive constant) $\int_{-\infty}^{\infty} \lvert X(j\omega)\rvert^2 \, d\omega = 2\pi$ Then the value of A is ____ .
Correct	3.46 (3.40-3.51)
Explanation	Given, $F^{-1}\{(1+j\omega)X(j\omega)\} = Ae^{-2t}u(t)$ $(1+j\omega)X(j\omega) = FT\{Ae^{-2t}u(t)\}$ $(1+j\omega)X(j\omega) = \frac{A}{2+j\omega}$ $X(j\omega) = \frac{A}{(1+j\omega)(2+j\omega)}$ $X(j\omega) = \frac{A}{(1+j\omega)(2+j\omega)} = A\left[\frac{A_1}{1+j\omega} + \frac{B_1}{2+j\omega}\right]$ $\therefore \quad A_1 = \frac{1}{2+j\omega}\Big\rvert_{j\omega=-1} = \frac{1}{2-1} = 1$

`

	$B_1 = \frac{1}{1+j\omega}\Big\|_{j\omega=-2} = \frac{1}{1-2} = -1$ $X(j\omega) = A\left(\frac{1}{1+j\omega} - \frac{1}{2+j\omega}\right)$ By taking inverse Fourier transform, $x(t) = Ae^{-t}u(t) - Ae^{-2t}u(t)$ From the fact-2, (or) $\frac{1}{2\pi}\int_{-\infty}^{\infty} \|X(j\omega)\|^2 \, d\omega = 1 = \int_{-\infty}^{\infty} \|X(t)\|^2 \, dt$ $\int_0^{\infty} \left\|Ae^{-t} - Ae^{-2t}\right\|^2 dt = 1$ $\int_0^{\infty} \left(A^2e^{-2t} - 2A^2e^{-3t} + A^2e^{-4t}\right)dt = 1$ $\Rightarrow \frac{A^2}{-2}\left(e^{-2t}\right)_0^{\infty} - \left(\frac{2A^2}{-3}\right)\left(e^{-3t}\right)_0^{\infty} + \left(\frac{A^2}{-4}\right)\left(e^{-4t}\right)_0^{\infty} = 1$ $\Rightarrow \quad \frac{A^2}{2} - \frac{2A^2}{3} + \frac{A^2}{4} = 1$ $\frac{6A^2 - 8A^2 + 3A^2}{12} = 1$ $\Rightarrow \quad \frac{A^2}{12} = 1$ $\Rightarrow \quad A = \sqrt{12} = 3.46$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

Question	614
Type	NAT
Body	Let $X(z)$ is the z-transformer of $x[n]$ and $X(z)=\frac{4z^2+2z}{2z^2-3z+1}$. Then the value of $x[n]$ at $n=0$ is ____ .
Correct	2(1 - 3)
Explanation	Given, $X(z)=\frac{4z^2+2z}{2z^2-3z+1}=\frac{2z(2z+1)}{2z^2-3z+1}$ $\frac{X(z)}{z}=\frac{2(2z+1)}{(2z-1)(z-1)}$ $\frac{X(z)}{z}=\frac{2(2z+1)}{(2z-1)(z-1)}$ $\frac{2(2z+1)}{(2z-1)(z-1)}=\frac{A}{2z-1}+\frac{B}{z-1}$ $A=2(2z+1)\|_{z=\frac{1}{2}}=\frac{2\left(2\times\frac{1}{2}+1\right)}{\frac{1}{2}-1}=-8$ $B=\frac{2(2z+1)}{2z-1}\|_{z=1}=\frac{2(3)}{1}=6$ $\therefore \quad \frac{X(z)}{z}=\frac{-8}{2z-1}+\frac{6}{z-1}$

`

	$X(z)=\frac{-8z}{2z-1}+\frac{6z}{z-1}=\frac{-4z}{z-\frac{1}{2}}+\frac{6z}{z-1}$ by taking inverse z-transformer, $x(n)=-4\left(\frac{1}{2}\right)^n u(n)+6u(n)$ at $n=0$, $x(0)=-4\times1+6=2$ Alternate Solution: Using the initial value theorem, $x(0)=\lim_{z\to\infty}X(z)$ $x(0)=\lim_{z\to\infty}\frac{2\left(2z^2+z\right)}{2z^2-3z+1}$ $=\lim_{z\to\infty}\frac{2\left(2+\frac{1}{z}\right)}{2-\frac{3}{z}+\frac{1}{z^2}}=\frac{2\times2}{2}=2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	615

`

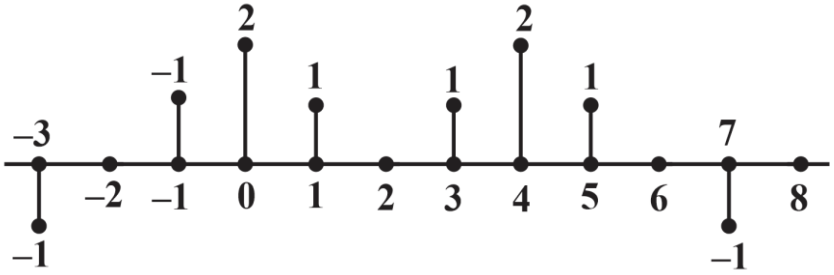
Type	NAT
Body	The discrete time Fourier transform of a particular discrete time signal $x[n]$ is $X(e^{j\omega})$ and $X(e^{j\omega}) = \sum_{k=0}^{3} \frac{\left(\frac{1}{2}\right)^k}{1-\frac{1}{4}e^{-j\left(\omega-\frac{\pi}{2}k\right)}}$. It can be shown that $x[n] = g[n]q[n]$, where $g[n]$ is of the form an $u[n]$ and $q[n]$ is a periodic signal with period N. Then the value of $N \times a$ is _____ .
Correct	1 (1-1)
Explanation	Given, $X(e^{j\omega}) = \sum_{k=0}^{3} \frac{\left(\frac{1}{2}\right)^k}{1-\frac{1}{4}e^{-j\left(\omega-\frac{\pi}{2}k\right)}}$ The given DTFT of $x[n]$ can be written as $X(e^{j\omega}) = \frac{1}{2\pi}\left\{\underbrace{\frac{1}{1-\frac{1}{4}e^{-j\omega}}}_{g[n]} * \underbrace{\left[2\pi\sum_{K=0}^{3}\delta\left(\omega-\frac{\pi}{2}K\right)\right]}_{q[n]}\right\}$ $\therefore \quad G(e^{j\omega}) = \frac{1}{1-\frac{1}{4}e^{-j\omega}} = \left(\frac{1}{4}\right)^n u[n]$ by comparing $g[n] = \alpha^n u[n]$ $\Rightarrow \quad \alpha = \frac{1}{4}$ $q[n]$ is a periodic with period N

`

	$\therefore \quad Q\left(e^{j\omega}\right)=2\pi\sum_{k=0}^{3}\delta\left(\omega-\frac{\pi}{2}k\right)$ For $0\le\omega\le 2\pi$ $Q\left(e^{j\omega}\right)=2\pi\left[\delta(\omega)+\delta\left(\omega-\frac{\pi}{2}\right)+\delta(\omega-\pi)+\delta\left(\omega-\frac{3\pi}{2}\right)\right]$ By taking inverse DTFT $q[n]=1+\frac{1}{2}e^{j\frac{\pi}{2}n}+\frac{1}{4}e^{j\pi n}+\frac{1}{8}e^{j\frac{3\pi}{2}n}$ $\therefore \quad$ Fundamental period, N = LCM (4, 2, 2) $\therefore \quad N=4$ Hence, $N\times\alpha=4\times\frac{1}{4}=1$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

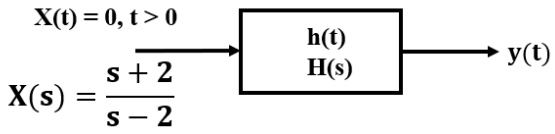
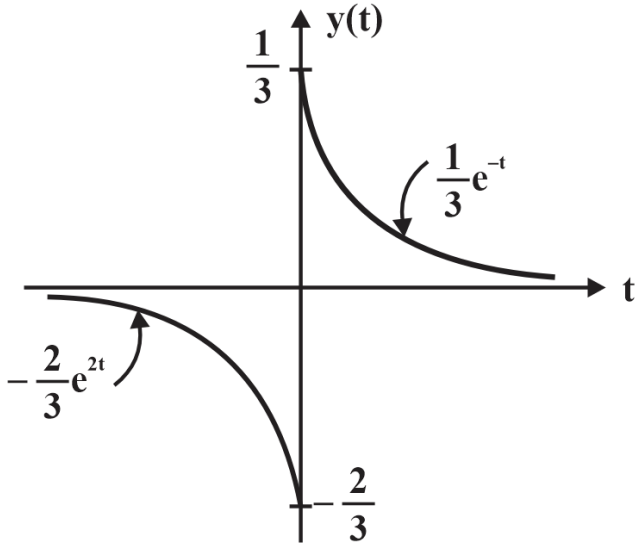
Question	616
Type	NAT
Body	Let $X\left(e^{j\omega}\right)$ denote the Fourier transform of the signal $x[n]$ shown in figure below,

`

	 Then the value of $\int_{-\pi}^{\pi}\left\|\frac{dX\left(e^{j\omega}\right)}{d\omega}\right\|^2 d\omega$ is __. (Answer upto two decimal places)
Correct	992.74 (990.00-995.00)
Explanation	Given, $\int_{-\pi}^{\pi}\left\|\frac{dX\left(e^{j\omega}\right)}{d\omega}\right\|^2 d\omega$ Using parseval's theorem, $\frac{1}{2\pi}\int_{-\pi}^{\pi}\left\|X\left(e^{j\omega}\right)\right\|^2 d\omega=\sum_{n=-\infty}^{\infty}\|x[n]\|^2$ From frequency differentiation property of DTFT, $nx[n]\xleftrightarrow{FT} j\frac{dX\left(e^{j\omega}\right)}{d\omega}$ $\therefore \quad \int_{-\pi}^{\pi}\left\|\frac{dX\left(e^{j\omega}\right)}{d\omega}\right\|^2 d\omega=2\pi\sum_{n=-\infty}^{\infty}\|nx[n]\|^2$ $=2\pi\left[\sum_{n=-\infty}^{\infty}\|n\|^2\|x[n]\|^2\right]$ $=2\pi[9\times1+4\times0+1\times1+0+1\times1+2\times0+9\times1+16\times4+25\times1$ $+36\times0+49\times1]$ $=316\pi$ $=992.74$
Subject	Signals and Systems
Topic	

`

Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	617
Type	MCQ
Body	Consider the LTI system shown below for which the following information is given,  $X(s) = \frac{s+2}{s-2}$ where,  The transfer function $H(s)$ is equal to

`

Option	$\frac{s}{(s-1)(s-2)};\text{Re}\{s\}<-2$
Option	$\frac{1}{(s+1)(s+2)};\text{Re}\{s\}>-2$
Option	$\frac{s}{(s+1)(s+2)};\text{Re}\{s\}>-1$
Option	$\frac{1}{(s+1)(s+2)};\text{Re}\{s\}>-1$
Correct	3
Explanation	Given, $X(s)=\frac{s+2}{s-2}$ $x(t)=0,t>0$ Output of LTI system can be written as $y(t)=-\frac{2}{3}e^{2t}u(-t)+\frac{1}{3}e^{-t}u(t)$ $Y(s)=\frac{\frac{2}{3}}{s-2}+\frac{\frac{1}{3}}{s+1}=\frac{s}{(s-2)(s+1)}$ ROC: $-1<\text{Re}\{s\}<2$ ROC of $X(s)$ is $\text{Re}\{s\}<2$ since it is given $x(t)$ is left sided signal $\therefore H(s)=\frac{Y(s)}{X(s)}=\frac{s}{(s-2)(s+1)}\times\frac{s-2}{s+2}=\frac{s}{(s+1)(s+2)}$ Since ROC of $Y(s)$ has to be the intersection of the ROC of $X(s)$ and $H(s)$ $\therefore$ ROC of $H(s)$ is $\text{Re}[s]>-1$ $\therefore H(s)=\frac{s}{(s+1)(s+2)};\text{Re}\{s\}>-1$

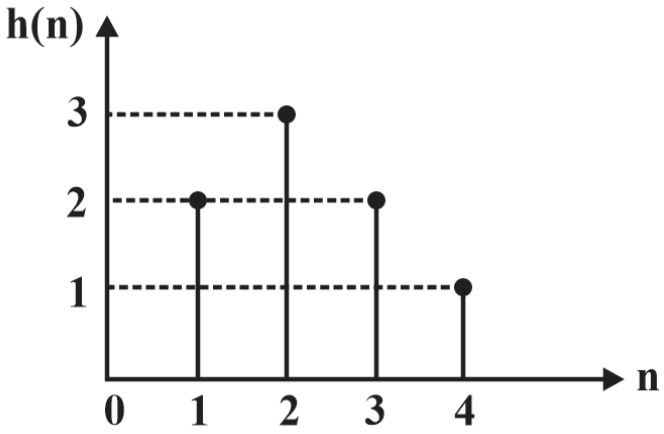
`

Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	618
Type	MCQ
Body	The Fourier transform $F(j\omega)$ of an arbitrary real signal has the property,
Option	$F(j\omega) = F(-j\omega)$
Option	$F(j\omega) = -F(-j\omega)$
Option	$F(j\omega) = F^*(-j\omega)$
Option	$F(j\omega) = -F^*(-j\omega)$
Correct	3
Explanation	If, $f(t) \xrightleftharpoons{F.T} F(j\omega)$ Then, $f^*(t) \xrightleftharpoons{F.T.} F^*(-j\omega)$ ...using conjugate property For any real signal,

`

	$f(t) = f^*(t)$ $\downarrow\uparrow$ F.T. $F(j\omega) = F^*(-j\omega)$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	619
Type	NAT
Body	Consider a filter with impulse response as shown in figure:  The group delay of filter is ____ .

`

Correct	2
Explanation	$h(n)=\delta(n)+2\delta(n-1)+3\delta(n-2)+2\delta(n-3)+\delta(n-4)$ $\downarrow\uparrow$ D.T.F.T. $H(e^{j\omega})=1+2e^{-j\omega}+3e^{-j2\omega}+2e^{-j\omega\omega}+e^{-j4\omega}$ $=\left[1+e^{-j4\omega}\right]+2\left[e^{-j\omega}+e^{-j3\omega}\right]+3e^{-j2\omega}$ $=e^{-2j\omega}\left[e^{j2\omega}+e^{-j2\omega}\right]+2e^{-j2\omega}\left[e^{j\omega}+e^{-j\omega}\right]+3e^{-j2\omega}$ $H(e^{j\omega})=e^{-j2\omega}\left[2\cos 2\omega+4\cos\omega+3\right]$ $=2\cos 2\omega+4\cos\omega+3$ $\angle H(e^{j\omega})=-2\omega$ Group delay is delay is, $\tau_g=\frac{-d}{d\omega}\left(\angle H\left(e^{j\omega}\right)\right)=2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	620
Type	NAT

`

Body	The z-transform $X(z)$ of a sequence $x(t)$ given by $X(z)=\frac{4z^{-1}}{1-2z^{-1}}$. It is given that, this region of convergence of $X(z)$ includes the unit circle. The value of $x(0)$ is ____ .
Correct	–2
Explanation	$\text{ROC}\lvert z\rvert<2$ (ROC should include unit-circle) $\therefore x(n)$ will be a left sided signal Now, $-(2)^n u(-n-1)\rightleftharpoons\frac{1}{1-2z^{-1}},\quad \lvert z\rvert<2$ $-4\times(2)^n u(-n-1)\rightleftharpoons\frac{4}{1-2z^{-1}}\quad \lvert z\rvert<2$ $-4\times(2)^{n-1} u(-n)\rightleftharpoons\frac{4z^{-1}}{1-2z^{-1}},\quad \lvert z\rvert<2$ So, $x(n)=-4(2)^{n-1}u(-n)$ $\therefore\quad x(0)=-4\left(2^{-1}\right)u(0)$ $=\frac{-4}{2}=-2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	

`

OTS ID	

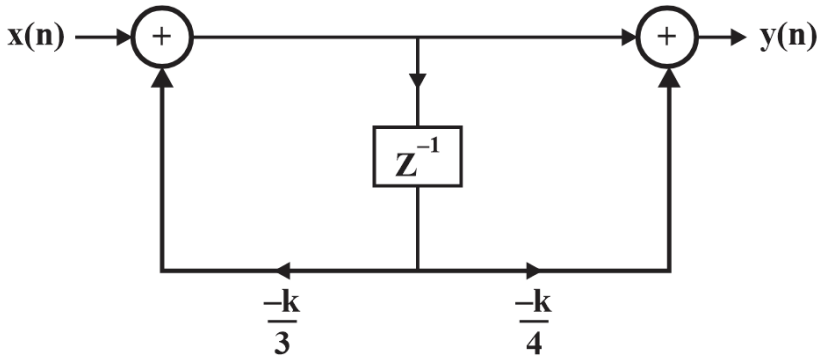
Question	621
Type	MCQ
Body	Suppose the following facts are given about the signal $x(t)$ with Laplace transform $X(s)$: 1. $x(t)$ is real and even. 2. $X(s)$ has four poles, no zeros in a finite s-plane. 3. $X(s)$ has a pole at $s = \frac{1}{2}e^{j\pi/4}$. 4. $X(0) = 4$ Then the Laplace transform $X(s)$ is
Option	$\frac{0.25}{s^4 + \frac{1}{16}}$
Option	$\frac{5}{s^3 + \frac{1}{4}}$
Option	$\frac{2}{s^2 + 4}$
Option	$\frac{4}{s^2 + 4}$
Correct	1
Explanation	From information-1, since $x(t)$ is real, there poles will occurs in conjugate pair,

`

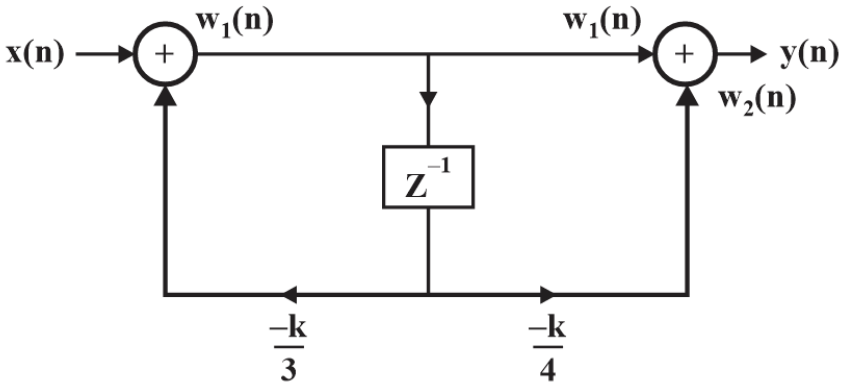
	$P_2 = P_1^* = \frac{1}{2}e^{j\pi/4}$ It is also given that $x(t)$ is even signal, i.e., $x(t) = X(-s)$, this implies that poles are negative of each other, Now, we can relates the poles as $P_3 = -P_1 \qquad P_4 = -P_2$ $P_3 = -\frac{1}{2}e^{j\pi/4} \quad P_4 = -\frac{1}{2}e^{j\pi/4}$ Now, $x(s) = \frac{K}{(s-P_1)(s-P_2)(s-P_3)(s-P_4)} = \frac{K}{(s^2-P_1^2)(s^2-P_2^2)}$ Also, $X(0) = 4$ $\frac{K}{(P_1^2 \cdot P_2^2)} = 4$ $K = 4(P_1P_2)^2 = 4 \times \left(\frac{1}{2} \times \frac{1}{2}\right)^2 = \frac{4}{16} = \frac{1}{4} = 0.25$ Therefore, $X(s) = \frac{0.25}{\left(s^2 - \frac{1}{4}e^{j\pi/2}\right)\left(s^2 - \frac{1}{4}e^{-j\pi/2}\right)} = \frac{0.25}{\left(s^4 + \frac{1}{16}\right)}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-

`

PYQ	
OTS ID	

Question	622
Type	MCQ
Body	Consider the causal digital filter shown in below figure:  The values of k for which system is stable.
Option	$\lvert k \rvert < 3$
Option	$\lvert k \rvert > 3$
Option	$\lvert k \rvert < \frac{1}{3}$
Option	$\lvert k \rvert > \frac{1}{3}$
Correct	1

`

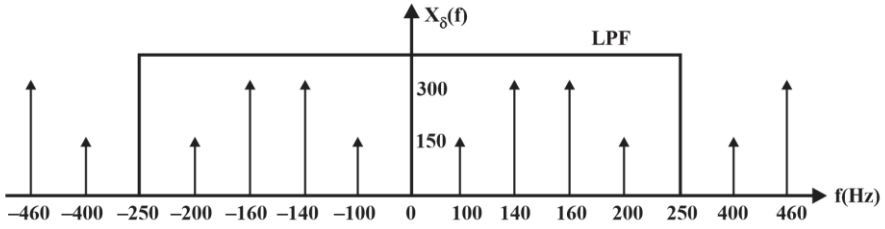
Explanation	 We have, $W_1(z) = X(z) - \frac{k}{3} z^{-1} W_1(z)$ $W_1(z) = \frac{X(z)}{1 + \frac{k}{3} z^{-1}}$ Also, $W_2(z) = -\frac{k}{4} z^{-1} W_1(z)$ $W_2(z) = -\frac{X(z)\frac{k}{4} z^{-1}}{1 + \frac{k}{3} z^{-1}}$ Therefore, $Y(z) = W_1(z) + W_2(z)$ will be $Y(z) = \frac{X(z)}{1 + \frac{k}{3} z^{-1}} - \frac{\frac{k}{4} z^{-1} X(z)}{1 + \frac{k}{3} z^{-1}}$ $H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{k}{4} z^{-1}}{1 + \frac{k}{3} z^{-1}}$ Since $H(z)$ corresponds to a causal filter, the ROC will be $\lvert z \rvert > \frac{\lvert k \rvert}{3}$. For the system to be stable, the ROC of $H(z)$

`

	must include the unit circle. This is possible only if $\frac{\left\|k\right\|}{3}<1$. $\therefore \left\|k\right\|<3$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	623
Type	MCQ
Body	A signal $x(t)=\cos(200\pi t)+2\cos(320\pi t)$ is ideally sampled at 300 Hz. If the sampled signal is passed through an ideal low-pass filter with cut-off frequency of 250 Hz. Highest frequency component will appear in the output is ____ Hz.
Correct	200
Explanation	This Fourier transform $X(f)$ of the given signal $x(t)$ is $X(f)=\frac{1}{2}\left[\delta(f-100)+\delta(f+100)\right]+\left[\delta(f-160)+\delta(f+160)\right]$

`

	With sampling frequency $f_s = 300$ Hz, the Fourier transform of the sampled signal is given as $X_\delta(f) = f_s \sum_{n=-}^{\infty} X(f - nf_s) = 300 \sum_{n=0}^{\infty} X(f - 300n)$ $X_\delta(f) = \ldots\ldots + 300X(f+300) + 300X(f) + 300X(f-300) + \ldots\ldots$ $X_\delta(f) = \ldots + 150\delta(f+200) + 150\delta(f+400) + 300\delta(f+140) + 300\delta(f+460)$ $+150\delta(f-100) + 150\delta(f+100) + 300\delta(f-160) + 300\delta(f+160)$ $+150\delta(f-400) + 150\delta(f-200) + 300\delta(f-460) + 300\delta(f-140) + \ldots..$ The sampled signal spectrum $X_\delta(f)$ is shown in figure below.  The frequency components that will appear at the output of the low-pass filter are 100,140,160 and 200 Hz. ∴ The highest frequency component present at the output is 200 Hz.
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

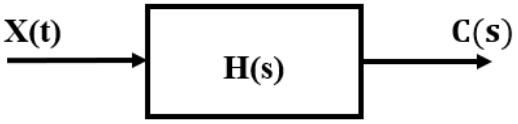
Question	624
Type	MCQ
Body	Consider the z-transform $X(z)=\frac{10-\delta z^{-1}}{2-5z^{-1}+2z^{-2}}$. If ROC includes unit circle then the value of $x(1)$ is
Option	0.5
Option	1
Option	1.5
Option	2
Correct	1
Explanation	Given, $X(z)=\frac{10-8z^{-1}}{2-5z^{-1}+2z^{-2}}=\frac{2}{\left(2-z^{-1}\right)}+\frac{4}{\left(1-2z^{-1}\right)}$ $X(z)=\frac{2z}{2z-1}+\frac{4z}{z-2}$ $X(z)=\frac{z}{\left(z-\frac{1}{2}\right)}+\frac{4z}{(z-2)}$ Since, ROC includes unit circle, $\therefore$ ROC of $X(z)$ is $\frac{1}{2}<\lvert z\rvert<2$ $x[n]=\left(\frac{1}{2}\right)^{n}u[n]-4\left(2^{n}\right)u[-n-1]$ $\therefore \quad x(1)=\frac{1}{2}=0.5$
Subject	Signals and Systems
Topic	

`

Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

Question	625
Type	MCQ
Body	The unit step response of a system with transfer function $\frac{b(s+a)}{(s+b)}$ is $c(t)$. If initial value $c(0)=2$ and final value $c(\infty)=8$. Then the ratio of $\left(\frac{a}{b}\right)$ is
Option	1
Option	2
Option	3
Option	4
Correct	4
Explanation	Let, the given transfer function, $H(s)=\frac{b(s+a)}{(s+b)}$

`

	 where, $X(s)=\frac{1}{s}$ ($\because$ unit step response) $C(s)=X(s)H(s)=\frac{b(s+a)}{s(s+b)}$ Given, $c(0)=2$ $\Rightarrow \quad Lt_{s\to\infty} s\cdot\frac{b(s+a)}{s(s+b)} = 2$ $b = 2$ and $c(\infty)=8$ $\Rightarrow \quad Lt_{s\to 0} s\cdot\frac{b(s+a)}{s(s+b)} = 8$ $a = 8$ now, $\frac{a}{b}=\frac{8}{2}$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	

`

Question	626
Type	MCQ
Body	For a discrete time LTI system, the input signal $x(n)=\{1,2,3,4\}$ and the output signal $y(n)=\{3,6,9,12\}$. If the unit impulse response of the given system is $h(n)$, then the value of summation $\sum_{n=-\infty}^{\infty} x(n)h(n)$ will be ____ .
Correct	0
Explanation	We can relate $y(n)$ and $x(n)$ as $y(n)=3x(n-2)$ $=x(n)*3\delta(n-2)$ $=x(n)*h(n)$ so, $h(n)=3\delta(n-2)$ $\sum_{n=-\infty}^{\infty} x(n)h(n)=\sum_{n=-\infty}^{\infty} x(n)\cdot 3\delta(n-2)$ $=3x(2)=0$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0
Hint	-
Video Solution	-
PYQ	
OTS ID	